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PASCAL: Pressure Adaptive Signal Control ALgorithm

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ABSTRACT

Max-pressure (MP) control has gained considerable attention in recent years, with numerous variations proposed. Existing MP studies primarily consider phase-based logic and are broadly classified into acyclic and cyclic approaches. Acyclic MP control often fails to ensure fairness in serving all movements, while cyclic MP control, with its fixed sequence of phases, restricts the prioritization of critical movements that require immediate attention. To the best of our knowledge, no MP control study has incorporated group-based control logic into the framework. Besides, most seminal and subsequent MP studies do not collectively address three practical challenges: (a) the growing number of vehicles in the candidate lane(s) of a movement, (b) reduced storage capacity in the downstream link, and (c) reduced storage capacity in the candidate lane(s). Some studies completely overlook capacity considerations, while others focus only on downstream spillover. A few incorporate storage capacities of both candidate lane(s) and downstream links into the pressure functions, but these fail to explicitly capture the growing demand in the candidate lane(s). Overall, no study holistically incorporates all these factors within the MP framework. To address these gaps, this study proposes a Pressure Adaptive Signal Control ALgorithm (PASCAL), which integrates group-based signal control logic with a novel pressure function that addresses the aforementioned challenges. To overcome the limitations of both acyclic and cyclic MP controls, PASCAL adopts a dynamic time slot-based critical movement prioritization, ensuring fairness in serving all movements without relying on a fixed phase sequence. Fairness is maintained through a critical movement selection pool, where active movements are removed before the next selection. The time slot depends on the green time considering the existing occupancy (queue length) of the critical movement or the available storage capacity in its downstream link, while being constrained by minimum and maximum green durations. The group-based policy is implemented by activating a set of critical complementary movements that achieve the highest total pressure among the predefined sets associated with the critical movement. PASCAL was benchmarked against four MP control models and a fixed-time control with SIDRA-optimized parameters under multiple demand and control scenarios. The MP models were: (a) the acyclic MP control by Varaiya (2013b), (b) the cyclic MP control by Le et al. (2015), (c) the MP with cyclical structure by Levin et al. (2020), and (d) the capacity-aware MP control by Gregoire et al. (2014). Results show that PASCAL consistently outperforms all benchmarking models, particularly under congestion, reducing mean delay, queue length, and network occupancy while maintaining higher speeds.

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1. Introduction

Traffic-responsive signal controls analyse real-time traffic conditions to manage urban traffic effectively. Over the years, these systems have gained widespread recognition, leading to numerous studies in this field. Among various control approaches, max-pressure (MP) control has recently gained considerable attention in the literature (Wongpiromsarn et al., 2012, Varaiya, 2013b, & 2013a) for its ability to control the signal without prior knowledge of traffic demand while maximizing the throughput of each intersection. Its practical relevance has also been demonstrated in recent field applications (Zoabi and Haddad, 2025).

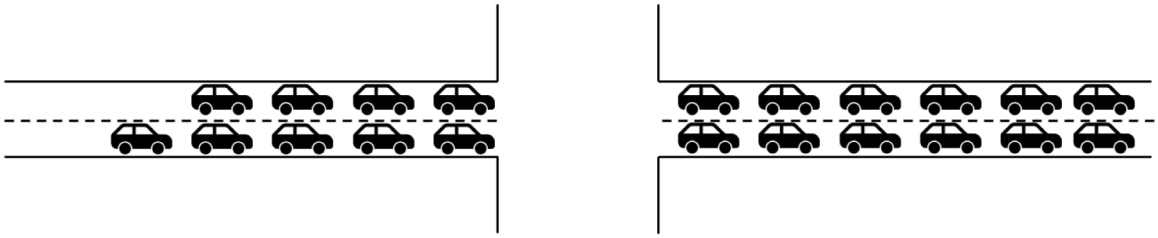
Fundamentally, the working principle of MP control is derived from the backpressure algorithm used in data packet transmission within wireless networks (Tassiulas and Ephremides, 1990). However, vehicles in a road traffic system and data packets in a wireless network behave differently, indicating that the backpressure data packet routing concept cannot be directly applied to the traffic networks. For instance, (a) many backpressure routing formulations assume packets share the same destination, whereas vehicles in a traffic network have diverse destinations; (b) traffic signal controllers cannot dictate vehicle routes, (c) signals influence rather than directly control link flows, as the actual flow depends on upstream congestion, driver behaviour, and downstream storage capacity; and (d) while backpressure routing often assumes infinite link storage, real traffic networks are constrained by finite link capacities (Wongpiromsarn et al., 2012).

MP control was first proposed for an isolated intersection by Wunderlich et al. (2007), and later extended to a traffic network by Wongpiromsarn et al. (2012). In parallel, MP control was also proposed by Varaiya (2013a & 2013b) for urban traffic networks. These seminal studies laid the groundwork for numerous subsequent studies covering various aspects of traffic control. MP controls use a pressure function to determine phase selection and phase timings. However, several practical challenges persist in existing MP control studies, particularly in the formulation of **phase selection criteria** (control logic) and **pressure function**.

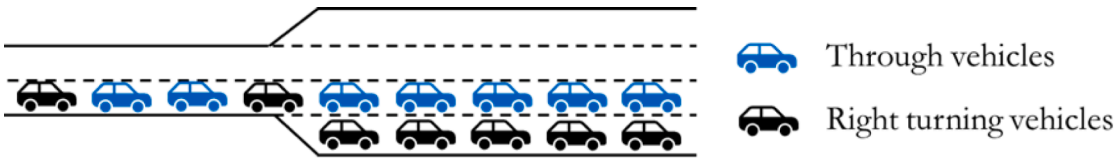
Traffic signals can be structured around two main control approaches: **phase-based** and **group-based**.

- In **phase-based control**, the signal operation is tied to a predefined set of non-conflicting movements (phases), which are activated either in an acyclic, time-step manner or in a fixed cyclic sequence. An acyclic approach allows any phase to be activated at each time step but risks skipping some phases while repeating others. A fixed cyclic approach guarantees coverage but lacks the ability to prioritize movements that require immediate attention. Although a few studies have introduced dynamic cycle times, they remain constrained by fixed sequences.
- In contrast, **group-based control** dynamically forms compatible sets of movements based on real-time traffic conditions. This approach ensures flexibility, supports fairness by serving all movements at least once, and allows green times to be responsive to demand and congestion levels. **All existing MP control models rely exclusively on phase-based logic, leaving group-based MP control largely unexplored.** Importantly, group-based logic has the potential to improve throughput by activating multiple non-conflicting movements simultaneously. Reflecting this potential, there is growing interest in Queensland, Australia, to incorporate group-based logic in their next generation traffic signal control program. Practitioners have shown strong interest in this paradigm, but it has not yet been incorporated into an MP framework. Such integration raises two novel challenges: (a) how to prioritize groups of movements under real-time traffic conditions, and (b) how to assign dynamic green durations once a group is selected.

The pressure function, which governs phase selection, has similar limitations. Standard formulations define pressure as the difference in occupancy (queue length) between upstream and downstream links. While effective in balancing flows, these formulations overlook several key practical considerations. For ease of discussion regarding the pressure function formulations, we define



(a) Lack of available storage capacity in the downstream link (**C2**)



(b) Lack of available storage capacity in the exclusive right turning lane in upstream link (**C3**)

Fig. 1. Selected practical challenges to be addressed by a signal control model.

downstream and upstream links in relation to the intersection controlled by the MP signal control model. Downstream links refer to the links that exit from the intersection, whereas upstream links (also termed as candidate links) are the links that enter the intersection. A practical pressure function should address the following three conditions:

- **Condition 1 (C1):** Longer queues in the upstream link should exert higher pressure.
- **Condition 2 (C2):** Reduced downstream spare capacity should reduce pressure to avoid wasted green time. Refer to Fig. 1(a), where a lack of downstream storage capacity can prevent dissipation of vehicles from upstream links, leading to wasted green time.
- **Condition 3 (C3):** Reduced upstream storage capacity (e.g., short turn bays) should increase pressure to prevent spillback. Refer to Fig. 1(b), where insufficient upstream storage capacity, particularly for exclusive right-turn lanes in left-hand driving networks, can cause blockages of through movements by right-turning vehicles.

Most existing models (Wongpiromsarn et al., 2012, Varaiya, 2013b, Le et al., 2015, Pumir et al., 2015, Xiao et al., 2015a, Xiao et al., 2015b, Anderson et al., 2018, Dixit et al., 2020, Levin et al., 2020, Ma et al., 2021, Zoabi and Haddad, 2022, Liu and Gayah, 2024) address **C1** only under special conditions. These models rely on the difference in occupancies between the candidate lane(s) and downstream links, therefore **C1** is satisfied only when downstream occupancies are identical across all movements at an intersection. A few studies (Gregoire et al., 2014, Kouvelas et al., 2014) partially capture **C2** and **C3**, but none incorporate all three factors in a unified formulation.

Following a comprehensive review of the existing MP control studies, as detailed in the next section, and the challenges explained earlier, it became evident that there exist three crucial attributes of urban network control that have not been explicitly considered in the existing MP studies. These are, (a) balanced approach to prioritize movements based on the number of vehicles and available spare capacities in the candidate link/ lane(s) and downstream link of a movement; (b) simultaneous activation of multiple safe movements to maximize the flow through the intersections; and (c) ensuring a fairness of serving all the movements at an intersection without relying on a fixed phase sequence. This paper addresses these gaps by introducing the **Pressure Adaptive Signal Control ALgorithm (PASCAL)**. It extends the MP framework through following innovations:

- Integration of a group-based policy within an MP framework, where the signal control identifies a group of complementary movements that are exerting the highest pressure and activates that group along with the identified critical movement.
- Formulation of a novel pressure function that considers not only the growing number of vehicles per movement but also the reduced storage capacity in both the candidate (upstream) and the downstream link of the movement. Moreover, it identifies the movement with the maximum pressure as the critical movement.
- Establishment of a fair process to ensure that all movements at an intersection are served at least once before being repeated as a critical movement by maintaining a dynamic Critical Movement Selection Pool (CMSP), which is updated at the end of each dynamic time slot. The model updates the CMSP by removing all movements that were served in the previous time slot and identifies the critical movement for the next time slot from the updated CMSP. Additionally, the time slot is dynamic, as it is determined by the green time required for the activated critical movements.

By combining the throughput-maximizing principles of MP with the flexibility of group-based control, PASCAL offers a practical and scalable framework for urban traffic signal management.

The remainder of the paper is organized as follows: Section 2 presents the detailed review of existing MP control studies. Section 3 outlines the methodology of the paper, starting with notations and network characteristics, followed by the formulation of the proposed pressure functions. Section 4 presents the numerical results, evaluation of PASCAL in a simulated traffic network, and a discussion of the findings. Finally, Section 5 concludes the paper and outlines the future work.

2. Literature review

Since the inception of MP control, a significant number of studies have been proposed with several modifications and applications to the original MP control. In the seminal MP control studies (Wongpiromsarn et al., 2012, Varaiya, 2013a), a signal phase is activated based on a pressure function, meaning the phase with the highest pressure is chosen at the beginning of each time step. The phase which is activated at the beginning of a decision time step is the critical phase. This pressure function depends on a weight function, which is typically defined for each movement. A phase, which consists of one or more movements, has a pressure function calculated as the sum of all movement weights multiplied by the saturation flow rate. Time is treated as discrete, with phase activation decisions made at regular intervals (say every 10 s). The weight function and the pressure function are recalculated at each decision step. Hence, with this method the green duration of a phase is incremented in every decision time step if the phase is identified as a critical phase repeatedly. This implies that such a method does not follow any specific sequence of phase activation and phases with lower pressure value may get skipped repeatedly. We name this method as **acyclic incremental green method** of MP. Over the years, a number of researchers (Xiao et al., 2014, Zaidi et al., 2016, Wu et al., 2018, Li and Jabari, 2019, Wei et al., 2019, Bai and Bai, 2021, Boukerche et al., 2021, Li et al., 2021, Liu and Gayah, 2022, Wang et al., 2022, Xu et al., 2022, Barman and Levin, 2023, Faqir et al., 2023, Liu and Gayah, 2023, Xu et al., 2024b, Gregoire et al., 2014) have adopted this method and incorporated changes in the weight and pressure function by addressing various aspects of signal control.

Another group of studies of MP control, adopted the traditional fixed sequence of phase activation, i.e., a cyclic structure, using the pressure function to calculate the green splits for the phases. We name this method as **cyclic MP control**. Majority of the literature of

cyclic MP control (Kouvelas et al., 2014, Le et al., 2015, Pumir et al., 2015, Xiao et al., 2015a, Xiao et al., 2015b, Anderson et al., 2018, Dixit et al., 2020, Mercader et al., 2020, Ma et al., 2021, Zoabi and Haddad, 2022, Tsitsokas et al., 2023) relied on fixed signal cycle. Limited researchers (Levin et al., 2020, Agarwal et al., 2024) retained a cyclic structure yet did not rely on a fixed cycle length. For instance, Levin et al. (2020) maintained a pre-defined phase sequence and adopted incremental green technique with a maximum cycle length constraint. Agarwal et al. (2024) retained the combinations of movements in each phase and adopted a dynamic phase sequence and phase timings technique.

A summary of the MP control studies is presented in Appendix A. The studies are clustered based on the abovementioned classifications. The performance measure, weight function, pressure function, phase selection criterion (for acyclic models) and green time calculation (for cyclic models) for each study are also presented with equations. The equations are presented in a simplified manner using notations for brevity. For consistency and ease of comparison, the formulations in Appendix A are expressed using unified notation aligned with that used in the present paper. In several cases, the pressure and weight functions from prior studies are reformulated or simplified to highlight their core structure. These representations reflect the authors' interpretation of the original methods, and readers are referred to the cited publications for the exact formulations, assumptions, and implementation details.

Wongpiromsarn et al. (2012) first adopted the MP control for an urban network taking the inspiration from internet packet routing (Tassiulas and Ephremides, 1990). Later on, Varaiya (2013a) adopted the MP control and analytically proved that the MP control is stabilizing. Maximizing throughput of an intersection is the main objective of an MP control. This is achieved by considering the downstream traffic congestion of each movement. The major difference in these two studies is the consideration of the queue length. In MP studies, queue lengths are not restricted to stopped vehicles only, rather, it is the occupancy of vehicles in a link. Wongpiromsarn et al. (2012) calculated the weight function as the difference in link-based queue lengths between the upstream and downstream links. On the other hand, Varaiya (2013a) considered movement-based queue length. In a link-based queue length MP system, for any turning movement, the entire link occupancy is considered. On the contrary, in a movement-based queue length system, only the occupancy of the specific turning movement is considered. For example, the occupancy of through movement and right turning movement is different. Hence, a link-based MP system may not be suitable for prioritizing a specific movement. Studies have used mostly the link/ lane(s) occupancies to incorporate various changes.

Two major issues with the seminal MP studies were (a) lack of fairness to activate all movements/ phases before repeating any; and (b) assumption of infinite capacity of links. The former limitation has already been discussed, as acyclic MP controls may repeatedly prioritize high-demand phases while neglecting lower-demand phases for a longer duration. Here, we discuss the latter limitation, which is also significant as the links have a finite storage capacities and spillover of links may lead to wastage of green time. For example, the spillover in turning bays (left and right turn lanes) leads to queuing of stopped vehicles in through movement lanes. Hence, the green duration for the through movement would be wasted as the through vehicles may not be able to dissipate due to spilled over turning vehicles. Similarly, if there is no remaining spare capacity in the downstream link of a movement, the vehicles from the upstream link cannot be accommodated at the downstream and the green time would be wasted. Xiao et al. (2014) and Gregoire et al. (2014) modified the weight/ pressure function by proposing a convex function of queue length and storage capacity. Below we present the limitations of these studies.

While the weight/ pressure function proposed by Gregoire et al. (2014) incorporates storage capacities, it does not explicitly capture the growing demand (queue length) of a candidate link/ lane(s). For instance, two conflicting movements may have the same ratio of queue length to storage capacity, since storage capacities depend on link length. However, the absolute queue lengths of the two movements can differ drastically. In such cases, even if both movements have adequate storage capacity in their respective downstream links, the weight/pressure function proposed by Gregoire et al. (2014) would not prioritize the movement with the higher queue length. Moreover, if the ratios of queue length to storage capacity for the candidate and downstream links are equal for a given movement, the weight function would evaluate to zero. In addition, Gregoire et al. (2014) utilized link-based queue lengths rather than movement-based queue lengths and it may not be suitable for the upstream turning bay spillover (Li and Jabari, 2019).

Xiao et al. (2014) considered the movement-based queue length along with the storage capacity in an acyclic MP system. The weight function proposed in their study essentially multiplies the sum of all further upstream queue lengths that could eventually join the candidate movement link. A potential shortcoming arises when high further-upstream queues do not actually arrive to the movement, since that depends on the signal timing at the upstream intersections. In other words, the over reliance on further upstream queue lengths is the limitation of this weight function. Furthermore, the optimal value of the parameter used in the weight function also influences the decision process. Additionally, the study does not provide any implementation and numerical results of the proposed weight/ pressure function.

Li and Jabari (2019) considered the spatial distribution of queue in a link and a weightage based MP formulation was devised in an acyclic system. However, sparse, or highly dense vehicle positions within a link or the potential absence of platoons, could result in inefficient signal control using this approach. This weight function assigns higher priority to vehicles closer to the stop line of a link, with gradually decreasing weight for vehicles further upstream. However, this approach might not accurately reflect the overall congestion patterns along the links. For example, consider two movements: one from a shorter link (say 150 m) with 10 vehicles densely packed near the stop line, and another from a comparatively longer link (say 250 m) with 20 vehicles sparsely distributed among various lanes in the link. Despite the longer link having higher congestion, the weight function may prioritize the vehicles in the shorter link due to the higher density of vehicles near stop line. Similar issues may arise during peak hours or high-demand conditions when all links are densely packed with vehicles. In such scenarios, the weight function might be overly influenced by minor variations in density between movements, rather than considering the actual remaining storage capacities of each movement, especially the turning movements with turning bays. This can result in inefficient prioritization of movements. Additionally, the bias of the weight function is due to the over reliance on vehicles near stop line of the links, which may not always correspond to the true congestion and

storage capacities along the entire link.

Yu et al. (2021) proposed two pressure equations: (a) endogenous pressure and (b) exogenous pressure. The endogenous pressure is the ratio of the demand for a movement to its allocated green time. The demand for a movement is defined as the difference between the estimated vehicles entering the movement and leaving it during the green time. The exogenous pressure expresses the ratio of available downstream storage capacity to the allocated green time for the movement. The limitation of the proposed pressure functions can be shown with an example. Consider a through and a right turning movement at an intersection with the same downstream storage capacity in their respective downstream links. Now, it is quite common for the through movement having higher demand but the right turning movement having a much smaller (or nearly full) storage capacity left in the dedicated right turning lane. Here, double pressure proposed by Yu et al. (2021), prioritizes the through movement having higher demand. However, ideally, the right turning movement should receive the green as it may already be overflowing into the through lane(s). In such a scenario, the vehicles in the through movement may not be able to dissipate the intersection even if through movement receives green. In other words, the method in Yu et al. (2021) may not explicitly account for cases where the upstream lane of a lower-demand movement is already near capacity.

Kouvelas et al. (2014) adopted a fixed sequence of phase activation and calculated green duration of a phase based on the ratio of pressure experienced by that phase to the total pressure exerted by all the phases. The study also considered finite capacity of links. However, the study relied on a link-based queue length. Le et al. (2015) also proposed a cyclic MP similar to Kouvelas et al. (2014). Nonetheless, the study did not incorporate lost times and minimum green times for phases. Hence, it may lead to a green time near zero if the pressure of a phase is much lower than other phases (Levin et al., 2020). Xiao et al. (2015b) and Xiao et al. (2015a) incorporated the lost time and minimum green time along with a cyclic structure. Pumir et al. (2015), Anderson et al. (2018) and Ma et al. (2021) calculated green times by solving an optimization problem. Ma et al. (2021) even incorporated variance of queue length in the optimization problem to consider the oversaturated traffic condition. However, these studies assumed infinite storage capacity of links. A major drawback of the fixed sequence cyclical structure is specific movement(s) cannot be prioritized despite the requirement of serving the movement(s) immediately. Additionally, majority of the cyclic MP studies relied on a fixed cycle time. Levin et al. (2020) addressed this issue of fixed cycle length by proposing a time-step based max-pressure control with dynamic cycle length. Each phase was activated at least once in each cycle with the constraints of minimum green time and maximum cycle length. The study solved a mixed integer linear program in a planning horizon to decide whether to continue with the current phase or move to the next phase in sequence. Nevertheless, the study did not consider the finite storage capacity of links. Moreover, the study considered a fixed phase sequence which limits the priority of activation of a specific movement.

In addition to queue lengths, several studies have proposed MP control using performance measures such as travel time, travel time delay, stopped delay, and average link delay. However, such performance measures may not reflect the storage capacity information of each movement accurately, particularly for turning movements. Wu et al. (2018) proposed two MP controls based on delay. The first control is solely based on delay where the weight function was calculated as the total time spent by the front row vehicles waiting at a link. The second control is a weighted model of delay and queue length. That said, the study assumed that the downstream link does not incur any delay, and the weight function is formulated only based on the upstream link. Furthermore, the study lacks the storage capacity consideration and cyclic structure. Dixit et al. (2020) used average link delay and vehicle volume to calculate the total delay for a turning movement. The study distributed the effective green time among a set of pre-defined phases based on the pressure function. Mercader et al. (2020) replaced queue length with the ratio of average travel time and free-flow travel time. This study also followed a fixed sequence of phases and distributed the green time based on the pressure function. This travel time metric can be intuitive for representing the storage capacity of lanes. Liu and Gayah (2022) proposed a MP control based on the delay of stopped vehicles in an acyclic MP system. Nevertheless, the delay of vehicles was considered as zero at the beginning of a time step even if the vehicles in a link are waiting for multiple time steps. In a later study, Liu and Gayah (2023) rectified this limitation by considering the total stopped delay of vehicles in a link. However, both the studies did not consider the storage capacity of links and both the studies followed an acyclic structure. Zoabi and Haddad (2022) formulated weight function using travel time delays and followed a cyclic structure with a constant cycle length.

A few studies have integrated vehicle route choice with MP control. For example, Zaidi et al. (2016) formulated the weight function using virtual link queues in an optimization problem. Liu et al. (2018) calculated weight function as the difference in queue lengths of upstream and downstream links associated with a destination index. The phase selection was based on the destination that maximizes the weight function. Other studies have combined reinforcement learning with MP control to enhance performance. For instance, Wei et al. (2019) and Boukerche et al. (2021) employed weight function to compute intersection pressure, and subsequently utilized the pressure function as the reward function of the reinforcement learning algorithm to optimize the signal. Wang et al. (2022) restricted the frequent switching of phases in an acyclic MP system to confine the loss time during phase switching. The study used reinforcement learning to optimize the weight function and switching policy. However, these studies lack finite storage capacity consideration and a cyclic structure. Agarwal et al. (2024) integrated the reinforcement learning algorithm with MP control in heterogeneous traffic condition using passenger car unit of each vehicle. The reinforcement learning model maintains a cyclic structure without relying on a fixed sequence. The algorithm optimizes the sequence and timings based on the estimated pressures of phases. Nevertheless, the weight function is same as Varaiya (2013b) and does not consider the storage capacities of the movements.

Several studies integrated MP formulations with additional traffic control measures. For instance, Barman and Levin (2023) explored the optimal locations of limited MP control deployment in an urban network considering budget constraint of implementing MP controls in an entire network. Xu et al. (2022) and Ahmed et al. (2024b) integrated transit signal priority and modified the MP phase selection by incorporating bus priority constraints. While Tsitsokas et al. (2023), Su et al. (2023) and Liu and Gayah (2024) combined perimeter control with MP control, Xu et al. (2024b) incorporated pedestrian phases into the traditional MP control. However, these studies do not explicitly consider the finite capacities of lanes and lack cyclic structure. Additionally, several studies

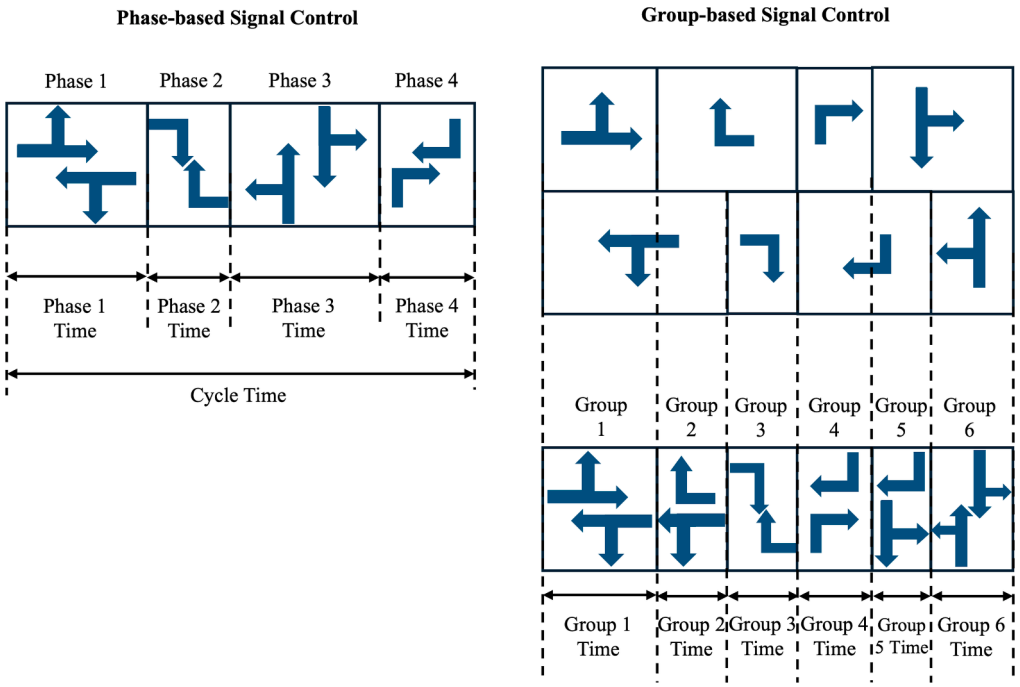


Fig. 2. A comparison of phase-based and group-based signal control (Figure depicts only vehicle movement groups, while pedestrian movements are omitted for brevity).

evaluated the performance of the MP controls through simulation experiments (Dakic et al., 2015, Sun and Yin, 2018, Sha and Chow, 2019, Wang and Abbas, 2019, Barman and Levin, 2022, Robbennolt et al., 2022). A comprehensive review of studies investigating max-pressure control, including detailed descriptions of each study, was conducted by Levin (2023). A few recent studies also integrated signal coordination with MP control. For instance, Xu et al. (2024a) added a coordination weight factor with the pressure function formulation of Varaiya (2013b). On the contrary, Ahmed et al. (2024a) incorporated a factor of average space mean speed and free-flow speed with the acyclic MP control weight function of queue length. In essence, the model gives higher priority to larger platoons in the upstream links and facilitates coordination of such platoons through intersections.

The review suggests that existing MP control studies are limited to incorporating all the necessary aspects of an urban traffic signal control. For instance, the existing methods often neglect to explicitly consider the movement-specific storage capacity in the lane(s) of the candidate link or for downstream link storage availability under increasing congestion. As per the authors' understandings, no study explicitly models the three major issues of a signal control combinedly, (a) growing congestion for a specific movement in the existing lane(s), (b) spilled over turning bay queues blocking other movements in the existing lane(s), and (c) spilled over downstream link due to lack of storage capacity. While some studies overlook storage capacity considerations entirely, others that address storage capacities do not explicitly account for demand. Growing congestion and reduced storage capacity in the candidate lane(s) necessitate the activation of the specific movement facing these issues. Conversely, if a particular movement has reduced storage capacity in the downstream link, the movement should not be activated, as the vehicles cannot be accommodated downstream. Therefore, in the present study, a novel pressure function is derived that provides a balanced approach considering the three issues mentioned earlier. The pressure function is used to identify a critical movement among a set of movements for each intersection in an urban network. To maximize the utilization of signal control and to serve multiple non-conflicting movements simultaneously, this study proposes a group-based signal control. This signal control identifies the critical complementary group of an identified critical movement and activates the group simultaneously. Additionally, many studies lack a cyclic structure, and the studies with a cyclic structure either lack capacity considerations or are confined to strict phase sequence. While an acyclic MP control leads to a lack of fairness to serve all the movements, a fixed sequence of phases limits the prioritization of a critical movement that requires immediate attention. To address the limitations of both the acyclic and cyclic MP controls, the present study adopted a dynamic time slot-based critical movement prioritization alongside a fair selection process of movement within a group-based framework, without relying on a fixed phase sequence. The fair selection process of critical movement is preserved by maintaining the CMSP, where active movements are replaced in the next time slot for critical movement identification. Additionally, the study incorporated minimum and maximum green time to ensure safety, efficient flow and reduce delay. Furthermore, amber time and clearance red time are provided to the activated movements.

Before discussing the detailed methodology of PASCAL, a distinction between stage (phase)-based and group-based signal control policy is necessary for comprehension. The major difference between them lies in the flexibility in both branching compatible movements (to form phases) and order of allocating green to the phases. In a phase-based framework, the phase structure and its

sequence are predefined. Here, a conflict matrix ensures the movements allocated to a phase are safe. In general, a signal cycle is maintained, and in adaptive systems there is flexibility in dynamic green time allocation and skipping phases with no demand.

In contrast, a group-based policy provides the highest level of flexibility with dynamic branching of compatible movements to be served concurrently, with no pre-defined phase sequence to be followed. Here, an inter-green matrix ensures the safe transition over different movements. Furthermore, a pre-defined cycle length is not required in a group-based signal policy. However, it is ensured that each movement receives a fair chance to be activated. Fig. 2 illustrates one possible phasing of a phase-based and a group-based signal policy at a typical four-legged intersection. Here, phase-based signal policy is divided into four distinct phases by batching multiple non-conflicting movements within each phase. These phases are activated in the fixed order as shown in the figure in every cycle. On the other hand, the figure on the right illustrates different non-conflicting groups that can be combined dynamically. The inter-green matrix specifies the minimum clearance time required for transitioning from one group to another. Additionally, certain groups may continue from one combination to the next, providing significant reduction in the loss time during phase transition. This dynamic grouping mechanism underscores the main contrast between the two strategies, highlighting how group-based control provides flexibility to be more responsive to changing traffic conditions.

With this background, Section 3 provides a detailed methodology of PASCAL signal control model.

3. Methodology

3.1. Network model

Consider an urban traffic network, indexed by $\mathscr{N}$. The network consists of multiple intersections, indexed as $\mathscr{J}$ and multiple road links, indexed as $\mathbb{Z}$. Hence, the network $\mathscr{N} = (\mathscr{J}, \mathbb{Z})$. Each intersection is denoted by $j \in \mathscr{J}$ and the set of road links $\mathbb{Z} = \mathbb{Z}_e \cup \mathbb{Z}_a \cup \mathbb{Z}_h$, where $\mathbb{Z}_e$, $\mathbb{Z}_a$ and $\mathbb{Z}_h$ are set of entry, internal and exit links in the network, respectively. Each intersection j has sets of entry and exit links denoted as $\mathscr{I}(j)$ and $\mathscr{O}(j)$, respectively. Each intersection consists of various movements $\mathscr{M}(j)$ and $\mathscr{M} = \mathscr{M}_{th} \cup \mathscr{M}_{lt} \cup \mathscr{M}_{rt}$, where $\mathscr{M}_{th}$, $\mathscr{M}_{lt}$, and $\mathscr{M}_{rt}$ are sets of through, left-turn and right-turn movements in an intersection. Each movement in an intersection is denoted by $m \in \mathscr{M}$ and each link $z \in \mathbb{Z}$ in an intersection belongs to one or more movements. Each movement is connected through a candidate link and a downstream link. Here, the candidate link is the link from which a movement originates. The movement m connecting these candidate and downstream links is denoted by the tuple (z_m, z_m^d), where z_m denotes one or more lane(s) in the candidate link z associated with movement m, and z_m^d denote the entire downstream link connected to movement m. Hence, the movement m can be denoted as, $m = \{(z_m, z_m^d), \forall z_m \in \mathscr{I}(j), \forall z_m^d \in \mathscr{O}(j)\}$. z_m^d belongs to the internal link of the downstream intersection, $z_m^d \in \mathscr{I}(j+1)$. In addition, for perimeter intersections, the exit links also act as z_m^d for certain movements.

In contrast to the traditional MP control models, which are phase-based, the proposed PASCAL is a group-based signal control model that first identifies a critical movement, denoted as $m_\Psi \in \mathscr{M}$, among the set of movements available in the critical movement selection pool (CMSP) and then activates a group of safe movements (referred to as critical complementary set) alongside it. For each movement, there can be multiple safe movements which can be activated individually with the critical movement. These are known as complementary movements, and each complementary movement is denoted as $m_\Theta \in \mathscr{M}$. However, all such complementary movements may not be activated together with the critical movement as some of the complementary movements are conflicting to each other. Hence, for each movement m, a collection of sets of complementary movements are identified. Let, $\mathbb{S}_m = \{\mathbb{S}_{m1}, \mathbb{S}_{m2}, \cdots, \mathbb{S}_{mn}\}$ be the collection of sets where each set $\mathbb{S}_{mi} \subseteq \mathscr{M} \backslash \{m\}$ represents a set of complementary movements that are safe with movement m. Movements in each complementary set are non-conflicting to each other and can be activated simultaneously. Out of all these complementary sets, the proposed model identifies a critical set that needs to be activated along with the critical movement m_Ψ. Such a set of complementary movements is named in this study as critical complementary movement set and denoted as $\mathbb{S}_{m_\Psi}$. The detailed description of this is provided in the subsequent section with an appropriate example.

Time is modelled as discrete steps, $t = 1, 2, \cdots$ and $\bar{t}$ is the time window which is the difference of two consecutive discrete time steps, $\bar{t} = (t+1) - t$. However, the signal control decision time step is not every discrete time step t. Instead, decision time step is a dynamic time slot which is an integer multiple of $\bar{t}$. At the end of a time slot, critical movement and the set of critical complementary movements are identified for activation in the subsequent time slot. Each dynamic time slot includes the green time g_m, for which the movement(s) are activated. Since PASCAL is a group-based signal control, the identification of a group of movements for the next time slot is performed before the amber time. This allows for the continuation of green for movements that are common between two consecutive time slots, avoiding unnecessary amber and clearance red times for such movements. Amber time Y and clearance red time $\widetilde{R}$ are then applied to movements that are not continued in the next time slot. All these durations can be represented as a function of the time window $\bar{t}$. For instance, $g_m = n_i \bar{t}$, $Y = n_Y \bar{t}$, and $\widetilde{R} = n_R \bar{t}$. Here, n_Y and n_R are constant integer multiples. However, n_i is a dynamic integer that represents the length of i-th dynamic green time for a group of movements at a specific intersection in the network. The representation of all the durations as discrete indices gives the flexibility to choose any time step size in the signal control algorithm. Therefore, the duration of each dynamic time slot can be denoted as $\overline{T}_i = n_i \bar{t}$ and in terms of the discrete time step, the i-th dynamic time slot can be written as $T_i = [t_1, t_1 + n_i \bar{t})$, where t_1 is the first discrete time step. Hence, the constant time gap between two successive time slots is the sum of amber and clearance red time. Note that discrete time step increases during the dynamic time slot, amber and clearance red time. The green time calculation for the critical movement m_Ψ and critical complementary set $\mathbb{S}_{m_\Psi}$ is performed at the beginning of the time slot T_{i+1} and these are activated until the end of T_{i+1}. The critical movement and the movements in the critical complementary set form a group for a specific time slot. As mentioned earlier, this group is identified at the end of the previous time

slot's green time, while the green time calculation for the group is performed at the beginning of the next time slot, i.e., end of the clearance red time. In other words, green time calculation is carried out at the time step (t) equals to $\left(T_i + Y + \widetilde{R}\right)$.

3.2. Vehicle propagation

The vehicle propagation model developed in this research aligns with the existing acyclic MP control approach and is based on the store-and-forward queuing model. The vehicle propagation model is presented with respect to the dynamic time slot, where the number of vehicles is estimated only at the end of a time slot when signal control decision is made for the next time slot, rather than at every discrete time step.

Let $x_{z_m z_m^d}(t+T_i)$ represent the occupancy or number of vehicles in the lane(s) of movement m of candidate link z, that are willing to move to the downstream link z_m^d connected to movement m observed at the beginning of time slot $t+T_i$. For brevity, the intersection index is omitted in the formulation, as the number of vehicles is calculated at a specific intersection in the network. Like the various other MP control studies, it is assumed that the information of number of vehicles associated with each movement is known.

Let $\mathbb{D}(\overline{T}_i)$ denote the demand for entry links $\mathbb{Z}_e$ during a time slot $\overline{T}_i$ and $\delta_m(\overline{T}_i)$ denotes the number of vehicles associated with movement m that can dissipate from the upstream link during the time slot $\overline{T}_i$ if the signal is activated. Here, δ_m is the saturation flow rate associated with the movement m. In this model, the lost time is compensated with the amber time and clearance red time given to the activated movement.

The number of vehicles served by a signal control policy during a time slot $\overline{T}_i$ corresponding to a movement m is defined as:

$$\mathscr{y}_{z_m z_m^d}(\overline{T}_i) = \delta_m(\overline{T}_i) S_m(t+T_i) \wedge x_{z_m z_m^d}(t+T_i) \tag{1}$$

where $\wedge$ denotes the minimum of the two elements. $S_m(t+T_i) \in \{1,0\}$ is a binary variable and $S_m(t+T_i)=1$ denotes that the signal for movement m is activated at the beginning of the time slot $t+T_i$. Equation (1) suggests that if the existing link/ lane(s) occupancy for a movement is less than the number of vehicles that can be dissipated from the link, all vehicles can move to the downstream link.

The vehicle propagation model is defined as follows:

$$x_{z_m z_m^d}(t+T_i) = x_{z_m z_m^d}(t) - \mathscr{y}_{z_m z_m^d}(\overline{T}_i) + \sum_{z_{m'} \in \mathscr{I}(j-1)} \mathscr{y}_{z_{m'} z_m}(\overline{T}_i) r_{z_m z_m^d}(\overline{T}_i) \forall z_m \in \mathbb{Z}_a \forall z_m^d \in \mathscr{O}(j) \tag{2}$$

$$x_{z_m z_m^d}(t+T_i) = x_{z_m z_m^d}(t) - \mathscr{y}_{z_m z_m^d}(\overline{T}_i) + \mathbb{D}_z(\overline{T}_i) r_{z_m z_m^d}(\overline{T}_i) \forall z_m \in \mathbb{Z}_e \forall z_m^d \in \mathscr{O}(j) \tag{3}$$

where $r_{z_m z_m^d}(\overline{T}_i)$ is the turning proportion of vehicles associated with movement m and $\mathbb{D}_z(\overline{T}_i)$ is the external demand entering link z. Equation (2) suggests that the number of vehicles at link z for a specific movement m at the beginning of the time slot $(t+T_i)$ equals to the number of vehicles observed at the beginning of the time step t subtracting the vehicles that leave the link during time slot $\overline{T}_i$ and adding the vehicles which are entering to the link z from further upstream links during the same time slot. Equation (3) is similar to Equation (2) but it is for the links which are connected to the perimeter and exogenous demand enters the link.

3.3. Fair selection of movements within a group-based framework

The fair selection of movements developed in this study neither follows a strict predefined sequence of movements, as in traditional cyclical MP studies, nor increments the green for any specific movement disregarding other movements for a longer period as seen in acyclic MP studies. As discussed earlier, both approaches have their own advantages and disadvantages. Failing to serve a specific movement for a longer duration leads to frustration and unsafe behaviour of drivers, while a fixed phase sequence restricts the signal control model to prioritize a critical movement. Therefore, in this study a critical movement selection pool (CMSP) is introduced to balance the trade-offs. Furthermore, PASCAL incorporates this fair selection process within the group-based framework.

Initially, at the beginning of the first time step ($t=0$), all movements at an intersection are included in CMSP. Note that, PASCAL only identifies the critical movement from CMSP. Once the critical movement is identified, the critical complementary set is selected from the predefined sets uniquely associated with the identified critical movement. As PASCAL serves these movements for the first dynamic time slot ($T_{i=1}$), they are removed from CMSP at the end of $T_{i=1}$. Consequently, the identification of the critical movement for the next dynamic time slot, T_{i+1}, is performed using the updated CMSP. This process continues until CMSP becomes empty, at which point all movements are reintroduced into CMSP. The critical movement is selected from CMSP with the constraint that the selected movement must have a positive pressure. If no critical movement is identified due to an empty CMSP or if all movements in CMSP have zero pressure, then CMSP is reset with all movements, and the selection process is repeated. Note that, in PASCAL model, the time index in each equation represents the end of the time slot. Consequently, any movement(s) identified by the signal policy at the end of a time slot is activated in the next time slot.

Let, at the end of the i-th dynamic time slot T_i for an intersection j the CMSP be $\mathfrak{P}_j$, that includes all the movements available for the intersection. This is defined as:

$$\mathfrak{P}_j(t+T_i) = \mathscr{M}(j) \tag{4}$$

Now the critical movement and critical complementary set identified based on the signal control policy at the end of T_i are $m_\Psi(T_i)$ and $\mathbb{S}_{m_\Psi}(T_i)$, respectively and can be represented as:

$$m_\Psi(t+T_i) \in \mathfrak{P}_j(t+T_i) \forall (t+T_i) \tag{5}$$

$$\mathbb{S}_{m_\Psi}(t+T_i) \subseteq \mathfrak{P}_j(t+T_i) \Leftrightarrow \left|\mathfrak{P}_j(t+T_i)\right| = |\mathscr{M}(j)| \tag{6}$$

Equation (5) ensures that the critical movement is within the CMSP for each time slot. Equation (6) suggests that the complementary set is within the CMSP only when the CMSP includes all the movements of an intersection. This is because one or more movements in a complementary set can be common to the movements activated in the last time step. In other words, complementary safe movements can be common for multiple critical movements. However, an entire complementary set cannot be identical to another complementary set. This is explained in the next section with a suitable example.

Now, the group of movements identified at the end of the time slot T_i, is activated at the beginning of the next time slot T_{i+1} except for the common movements which are already active. At the end of the time slot, T_{i+1}, CMSP replaces the movements m_Ψ and $\mathbb{S}_{m_\Psi}$. The selection of the critical movement and critical complementary set at the end of T_{i+1} is from this updated pool. Hence, the updated CMSP at the end of T_{i+1} can be defined as follows:

$$\mathfrak{P}_j(t+T_{i+1}) = \mathfrak{P}_j(t+T_i) \backslash (\{m_\Psi(t+T_i)\} \cup \{\mathbb{S}_{m_\Psi}(t+T_i)\}) \tag{7}$$

This process is continued until $\left|\mathfrak{P}_j\right| = 0$ and then all the movements are replaced in the pool.

3.4. PASCAL signal control policy

The major step in the proposed PASCAL signal control policy is the identification of critical and corresponding complementary movements set at the end of a dynamic time slot for the immediate next time slot. This is based on the following three major criteria which are the foundation of PASCAL:

1. Growing number of vehicles in the lane(s) of the movement;
2. The reduced storage capacity in the lane(s) of the movement; and
3. The reduced storage capacity in the downstream link of the movement.

These criteria are essential for efficient vehicle dispersion in signalized intersections and reduction of vehicular congestion. With the growing number of vehicles in the lane(s) of a movement, the delay for that specific movement as well as for the entire intersection increases. Therefore, it is important to prioritize a movement with a greater number of vehicles than other movements in an intersection. However, the spare capacity in the movement lane(s) is also essential as it may interrupt the dissipation of vehicles of other movements even if the queue length is higher in other movements. In this case, the spare capacity is the difference between the total number of vehicles that can be stored in the lane(s) allocated to a specific movement and the number of vehicles present in those lane (s). For example, the turning bays (left and right turn lanes) usually have much lower capacity than the through movements. If a turning bay has no spare capacity left, the turning vehicles are queued in through lane(s). In this scenario, even if the through movement has higher queue length than the turning movement, providing green signal to the through movement does not help the vehicles in through lane(s) to dissipate from the intersection and the green time is wasted. Moreover, if a specific movement lane has limited spare capacity, movement should be prioritized. Therefore, it is essential to consider the spare capacity of the movement lane along with the number of vehicles. With these two important considerations, another important criterion is the spare capacity in the downstream link of a movement. If there is no remaining spare capacity in the downstream of a movement, the vehicles from the upstream link cannot be accommodated at the downstream and the green time would be wasted. Note that, as mentioned earlier, the signal control model is proposed with respect to the dynamic time slot T_i instead of the discrete time step t.

The first criterion is fulfilled simply with the term $x_{z_m z_m^d}(t+T_i)$ that represents the queue length (number of vehicles) waiting at the lane(s) of link z associated with the movement m to move to the downstream link z_m^d at the end of the i-th dynamic time slot. The second criterion directly relates to the priority of activating a movement. Therefore, this criterion is quantified by the lane utilization factor, represented as $\frac{x_{z_m z_m^d}(t+T_i)}{c_{z_m}}$, where c_{z_m} is the storage capacity of the lane(s) of link z associated with the movement m. This storage capacity is represented in terms of number of vehicles that can be stored by the lane(s) when the vehicles are stopped. This storage capacity can be represented as:

$$c_{z_m} = \frac{\sum_z \xi_{z \in z_m}}{\overline{\mathscr{K}}} \tag{8}$$

where $\xi_{z \in z_m}$ is the length of a lane z that is associated with movement m in link z. $\overline{\mathscr{K}}$ is the average distance between the front end of a vehicle to the front end of the next vehicle when both the vehicles are stopped in a queue, i.e., inverse of jam density. The factor formulated for the second criterion increases as the number of vehicles increases, indicating the diminishing storage capacity in the movement lane(s). For example, if the turning bays are left with minimum storage capacity, this term will maximize the function, regardless of whether the through movement exhibits a higher queue length. Consequently, this term prioritizes the movements with

minimal storage capacity in the upstream of a movement.

The third criterion corresponds to reduced spare storage capacity in the downstream link associated with a movement. When the remaining storage capacity in the downstream link is limited, discharging additional vehicles from the upstream intersection may not result in effective flow, and allocated green time may be wasted. Consequently, movements that lead to downstream links with limited residual storage capacity should receive lower priority for activation. Note that, unlike the first and second criteria, which are defined for the lane(s) associated with a movement, the downstream storage capacity is evaluated at the link level, since the turning movement chosen by vehicles after entering the downstream link is not known at the time of decision-making. It is also worth mentioning that localised, movement-level spillback within the downstream link is addressed indirectly through the signal control at the downstream intersection, since the second criterion defined for the downstream intersection accounts for this effect.

This may be inferred that the third criterion is inversely related to the prioritization of movement activation. However, it cannot be directly expressed as the inverse of link utilization factor, i.e., $\left(\frac{1}{\frac{x_{z_m^d}(t+T_i)}{c_{z_m^d}}}\right)$. This is because there might be instances where no vehicles are present in the downstream link of a movement. In such cases, this function becomes undefined due to division by zero. Hence, the reduced spare capacity in downstream link is represented as $\left(1-\frac{x_{z_m^d}(t+T_i)}{c_{z_m^d}}\right)$. This term decreases as the storage capacity in the downstream link of a movement decreases. For instance, if the storage capacity of a movement in the downstream link decreases, this term assigns lower priority to that movement.

A composite pressure function is formulated by combining these three criteria functions to decide which movement should be activated at the end of the i-th dynamic time slot T_i. To formulate the pressure function, three functions are multiplied, allowing the influence of each individual function to be reflected in the pressure function. For instance, if the functions were added, and if there was no storage capacity in downstream link for a particular movement, only the third function $\left(1-\frac{x_{z_m^d}(t+T_i)}{c_{z_m^d}}\right)$ would become zero, yet the pressure function would still retain some value. However, in such a scenario, the pressure function should attain zero value, and the movement should not be activated. Thus, to ensure accurate representation, the functions are multiplied to form the pressure function. The pressure function $P_j^m(t+T_i)$ for movement m of intersection j at the end of i-th dynamic time slot is defined as follows:

$$P_j^m(t+T_i)=x_{z_m z_m^d}(t+T_i)\left(\frac{x_{z_m z_m^d}(t+T_i)}{c_{z_m}}\right)\left(1-\frac{x_{z_m^d}(t+T_i)}{c_{z_m^d}}\right)\Bigg|\forall m\in\mathscr{M}(j)\forall\left(z_m,z_m^d\right)\in\mathbb{Z}_a{}^2\forall j\in\mathscr{J} \tag{9}$$

$$P_j^m(t+T_i)=x_{z_m z_m^d}(t+T_i)\left(\frac{x_{z_m z_m^d}(t+T_i)}{c_{z_m}}\right)\Bigg|\forall m\in\mathscr{M}(j)\forall z_m\in\mathbb{Z}_a\forall z_m^d\in\mathbb{Z}_h\forall j\in\mathscr{J} \tag{10}$$

where Equation (10) is valid when the downstream link z_m^d is an exit link $\mathbb{Z}_h$, and the exit links have infinite storage capacity.

Remarks on boundedness and non-linear coupling:

The above-mentioned non-linear pressure function, $\frac{\left(x_{z_m z_m^d}(t)\right)^2}{c_{z_m}}\left(1-\frac{x_{z_m^d}(t)}{c_{z_m^d}}\right)$, can be rewritten in normalised form as, $c_{z_m}\left(\frac{x_{z_m z_m^d}(t)}{c_{z_m}}\right)^2\left(1-\frac{x_{z_m^d}(t)}{c_{z_m^d}}\right)$. The multiplicative structure of the proposed pressure function is intentionally designed as a bounded regulatory mechanism rather than an amplifying interaction. The upstream queue ratio, $\left(\frac{x_{z_m z_m^d}(t)}{c_{z_m}}\right)$, and the downstream storage availability ratio, $\left(1-\frac{x_{z_m^d}(t)}{c_{z_m^d}}\right)$, are both normalised by their respective finite physical link capacities and therefore remain bounded within the interval, [0,1].

This boundedness reflects the physical constraints of the traffic network. Once a link reaches its storage capacity, additional arriving demand manifests as spillback to upstream links or as virtual queues outside the network, rather than as further growth in the occupancy of the same link. Accordingly, the non-linear interaction among the three terms remains inherently self-regulated.

In particular, as the storage availability in the downstream link decreases, the corresponding regulatory term approaches zero, which suppresses the overall pressure irrespective of the upstream demand magnitude. This damping effect is particularly important under spillback conditions and prevents numerical amplification or unstable escalation of pressure values.

Furthermore, the squared demand term for the candidate lane(s), $\left(x_{z_m z_m^d}(t)\right)^2$, improves discrimination under moderate congestion. Its effect is dominant only when downstream storage availability is adequate, and its influence is explicitly damped under downstream congestion.

To further illustrate this behaviour, Fig. 3 presents the variation of the normalised pressure, $\left(\frac{P_j^m}{c_{z_m}}\right)$, with respect to the queue-to-

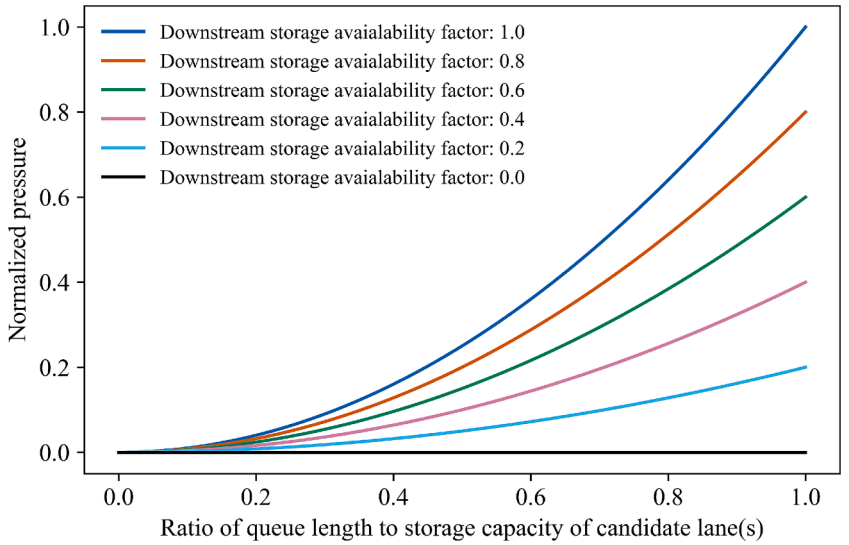
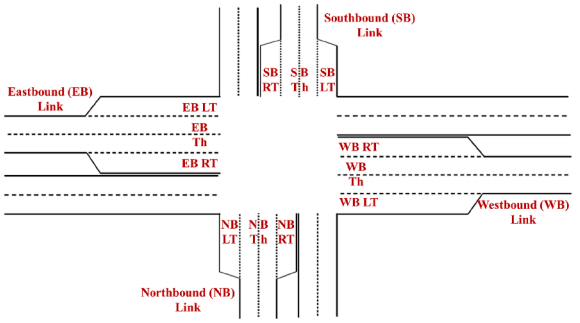


Fig. 3. Variation of the normalised pressure with respect to the queue-to-capacity ratio of the candidate lane(s) under different levels of downstream storage availability, illustrating bounded non-linear coupling behaviour.



	NB Th	NB LT	NB RT	SB Th	SB LT	SB RT	EB Th	EB LT	EB RT	WB Th	WB LT	WB RT
NB Th	-	1	1	1	1	0	0	0	0	0	1	0
NB LT	1	-	1	1	1	0	1	1	1	0	1	1
NB RT	1	1	-	0	0	1	0	1	0	0	1	0
SB Th	1	1	0	-	1	1	0	1	0	0	0	0
SB LT	1	1	0	1	-	1	0	1	1	1	1	1
SB RT	0	0	1	1	1	-	0	1	0	0	1	0
EB Th	0	1	0	0	0	0	-	1	1	1	1	0
EB LT	0	1	1	1	1	1	1	-	1	1	1	0
EB RT	0	1	0	0	1	0	1	1	-	0	0	1
WB Th	0	0	0	0	1	0	1	1	0	-	1	1
WB LT	1	1	1	0	1	1	1	1	0	1	-	1
WB RT	0	1	0	0	1	0	0	0	1	1	1	-

Fig. 4. Example of a four-legged intersection consisting of twelve movements (Left figure), and conflict matrix (Right figure) of these movements provided for ease of discussion.

capacity ratio, $\left(\frac{x_{z_m z_m^d}(t)}{c_{z_m}}\right)$, of the candidate lane(s) under different levels of downstream storage availability, $\left(1-\frac{x_{z_m^d}(t)}{c_{z_m^d}}\right)$. It can be observed that when the downstream link is uncongested, i.e., $\left(1-\frac{x_{z_m^d}(t)}{c_{z_m^d}}\right)=1$, the pressure is more sensitive to variations in the non-linear term. In contrast, when the downstream link approaches full occupancy, i.e., $\left(1-\frac{x_{z_m^d}(t)}{c_{z_m^d}}\right)=0$, the downstream storage term suppresses the pressure, thereby demonstrating that the non-linear interaction acts as a damping mechanism rather than an amplifier.

The pressure function is used to calculate the pressure for each movement of an intersection. Therefore, to decide the critical movement, the movement with maximum pressure is selected which is the MP policy. Hence, the MP policy at the end of $(t+T_i)$ time slot is:

$$S^*(t+T_i)=\underset{m\in\mathscr{M}(j)}{\operatorname{argmax}}\delta_m(t+T_i)P_j^m(t+T_i) \tag{11}$$

where Equation (11) results the critical movement $m_\Psi(t+T_i)$ at the end of time slot T_i.

The next step is to identify the set of critical complementary movements that should be prioritized and activated along with the critical movement. Prior to defining the critical complementary movements set, it is important to understand the conflict matrix of an intersection and complementary movements sets construction. This is explained with an example of a four-legged intersection along with the conflict matrix as presented in Fig. 4. In this example, the intersection has separate lane(s) for each movement, and one movement does not share lane with another movement. Hence, the intersection consists of twelve movements, i.e., three movements for each link. On the right side of the figure, the conflict matrix is shown where each movement is checked with respect to other

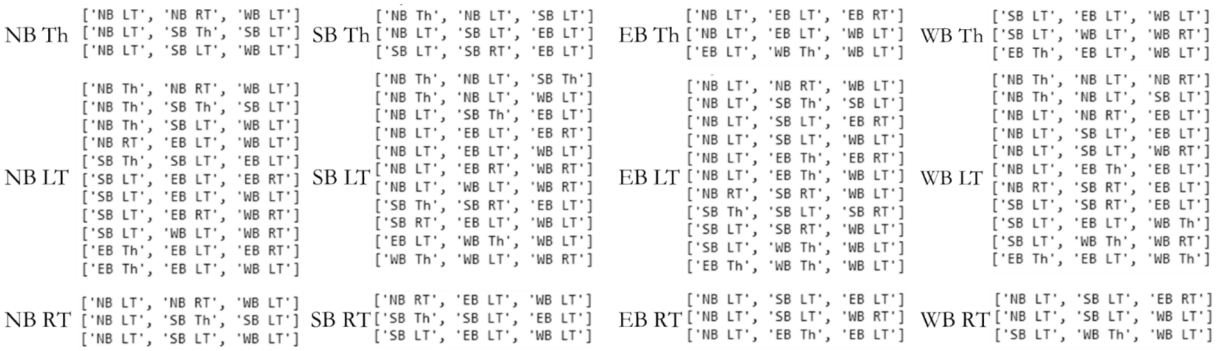


Fig. 5. Sets of complementary movements corresponding to each movement as shown in the previous example.

movements whether conflicting or non-conflicting. In the conflict matrix, 1 indicates that the movement in the column is safe and complementary with the movement in the row, while a value of 0 indicates otherwise.

Based on the conflict matrix shown in the example, it may be observed that each movement is non-conflicting with various movements. However, these movements can be conflicting to each other. For example, northbound through (NB Th) is non-conflicting with northbound right turn (NB RT) and southbound through (SB TH) individually. However, NB RT and SB TH are conflicting to each other. Hence, for each movement, multiple sets of complementary movements are formed. As mentioned earlier, within each set, multiple movements are grouped together, ensuring that the movements are non-conflicting to each other. Corresponding to the example above, the sets of complementary movements for each movement are shown in Fig. 5.

As each complementary movement set consists of multiple movements, pressure function for a set is calculated as the sum of the pressure functions for all the movements in a set. Hence, the pressure function for each set of complementary movements corresponding to the critical movement is defined as follows:

$$P_j^{\mathbb{S}_m}(t+T_i) = \sum_{m \in \mathbb{S}_m} \delta_m(t+T_i) P_j^m(t+T_i) \tag{12}$$

Now to identify the critical complementary set the policy is defined as:

$$S^*_{\mathbb{S}_m}(t+T_i) = \underset{\mathbb{S}_m \in \mathbb{S}}{\operatorname{argmax}} P_j^{\mathbb{S}_m}(t+T_i) \tag{13}$$

where Equation (13) identifies the critical complementary set $\mathbb{S}_{m_\Psi}(t+T_i)$ corresponding to the critical movement $m_\Psi(t+T_i)$ at the end of time slot $(t+T_i)$.

PASCAL model evaluates all the movements within a complementary movement set in each time step irrespective of whether any of the movements within a set is served in the previous time step or not. PASCAL takes signal control decision in two steps, (a) at first, the critical movement is selected, (b) then the critical complementary movements set is selected. The CMSP is updated only for the purpose of identifying the critical movement. Specifically, any movement that was served as a critical movement or was part of the critical complementary set in the previous time step is removed from the updated CMSP strictly for choosing the critical movement at the current time step.

Each movement has uniquely associated complementary sets, and once a particular movement is designated as critical, its critical complementary set is chosen from among these predefined sets. During this selection, the pressure of each complementary set is calculated by including all movements in that set, regardless of whether any of those movements were activated in the previous time step. For instance, at a four-legged intersection with no shared or slip lane, each leg consists of three turning movements, through (Th), right turn (RT) and left turn (LT). For instance, if Northbound Through (NB Th) is identified as the critical movement at a time step, three possible complementary sets may exist:

a) [NB LT, NB RT, WB LT],
b) [NB LT, SB Th, SB LT],
c) [NB LT, SB LT, WB LT].

The total pressure is computed for each set, and the one with the highest total pressure (say, set b) is then selected as the critical complementary set. All movements in that set are activated at that time step, even if one of them, such as SB Th, had been served previously as a critical movement or as part of another complementary set.

PASCAL incorporates dynamic green time allocation, which determines the duration for which the selected critical movement and critical complementary set of movements are activated. Note that this green time decides the duration of dynamic time slot. To calculate the green time, the time required for the number of vehicles present in the lane(s) associated with the critical movement is calculated. However, it is crucial to consider that the number of vehicles present in the upstream may not entirely be accommodated in the downstream link due to the downstream available storage capacity constraint. Therefore, the storage capacity of downstream link

is also incorporated in the green time estimation, $g_m\left(t+T_i+Y+\widetilde{R}\right)$. The green time is now defined as follows:

$$g^m\left(t+T_i+Y+\widetilde{R}\right)=\max\left(\min\left(\frac{x_{z_m z_m^d}\left(t+T_i+Y+\widetilde{R}\right)}{\delta_m},\frac{\left(c_{z_m^d}-x_{z_m^d}\left(t+T_i+Y+\widetilde{R}\right)\right)}{\delta_m},g^m_{max}\right),g^m_{min}\right) \tag{14}$$

where g^m_{min} and g^m_{max} are the constant minimum and maximum green durations, respectively. Note that, here, green time is represented with respect to the time step $\left(t+T_i+Y+\widetilde{R}\right)$, which marks the beginning of the dynamic time slot T_{i+1}. As mentioned earlier; to leverage the advantages of the group-based signal control model, the selection of critical movement and its corresponding critical complementary set is performed at the end of the time step $(t+T_i)$. This prevents unnecessary amber and clearance red times for movements that continue into the next time slot. However, the green time calculation for the identified critical movement and its critical complementary set is carried out at the end of the clearance red time or at the beginning of the next dynamic time slot T_{i+1}. The calculated green time is a measure of the next dynamic time slot T_{i+1} as follows:

$$\left|g^m\left(t+T_i+Y+\widetilde{R}\right)\right|=|T_{i+1}| \tag{15}$$

3.5. An example of PASCAL workflow

Suppose the time step t, begins at 6:00:00 AM. At this point, the CMSP includes every possible movement in the intersection.

$$\mathfrak{P}_j(t)=[NBTh,NBLT,NBRT,SBTh,SBLT,SBRT,EBTh,EBLT,EBRT,WBTh,WBLT,WBRT] \tag{16}$$

Now PASCAL computes the following steps:

a) Pressure calculation: PASCAL calculates the pressure value for each movement in the CMSP using the PASCAL pressure function.

b) Critical movement identification: Assume it identifies "Southbound Through (SB Th)" as the critical movement. This suggests the decision of the signal control model is denoted as follows:

$$S^*(t)=\underset{m\in\mathscr{M}(j)}{\operatorname{argmax}}P_j^m(t)=[\boldsymbol{SBTh}] \tag{17}$$

c) Critical complementary set identification: Next, PASCAL chooses one complementary set from among the pre-defined sets associated with "SB Th." The identified sets for all the movements are shown in Fig. 5. The figure shows there are three different complementary sets associated with SB Th. Each set is evaluated by summing the pressures of the movements in that set. Suppose the set with the highest sum of pressures is:

$$S^*_{\mathbb{S}_m}(t)=\underset{\mathbb{S}_m\in\mathbb{S}}{\operatorname{argmax}}P_j^{\mathbb{S}_m}(t)=[\boldsymbol{NBLT},\boldsymbol{SBLT},\boldsymbol{EBLT}] \tag{18}$$

Combining this with the critical movement, the group to be activated at the current time becomes: $[SBTh,NBLT,SBLT,EBLT]$.

d) Green time calculation: PASCAL determines the green time based on the requirements of the critical movement (SB Th). Let the calculated green time $g(t)$ be 25 s. This group is activated immediately for those 25 s.

e) Movement overlapping check after green time: Before the green interval for this group ends at 6:00:25 AM, PASCAL identifies the next critical movement and its complementary set. This step is necessary because any movement that overlaps in both the current and the next group does not require an amber or clearance red phase; its green is simply continued. At the end of the green time, i.e., 6:00:25 AM, when the current green ends, PASCAL updates the CMSP by removing the movements just activated:

$$\mathfrak{P}_j(t+g)=[NBTh,NBRT,SBRT,EBTh,EBRT,WBTh,WBLT,WBRT] \tag{19}$$

PASCAL recalculates pressures for these remaining movements. Suppose it identifies "NB Th" as the next critical movement, and the first complementary set (see Fig. 5) is selected as the critical complementary set, i.e., $[NBLT,NBRT,WBLT]$. Thus, the next group to be activated is $[NBTh,NBLT,NBRT,WBLT]$. In this example, [NB LT] is common to both groups, so it continues receiving green without interruption.

Similar to the previous step, green time is calculated for this group, g', say 30 s. However, this new green interval will start once the amber and clearance red periods end for the movements from the previous time slot, except for any movements common to both groups.

f) Amber and clearance red: Meanwhile, the other movements from the previous group, [SB Th, SB LT, EB LT], receive amber and clearance red times of 4 s. Specifically, at 6:00:25 AM, PASCAL initiates the amber phase for those movements. At 6:00:27 AM, it starts the clearance red phase. During these 4 s, [NB LT] remains green.

Finally, at 6:00:29 AM, the new group [NB Th, NB LT, NB RT, WB LT] begins its allocated green interval, and the signal control process continues in a similar manner.

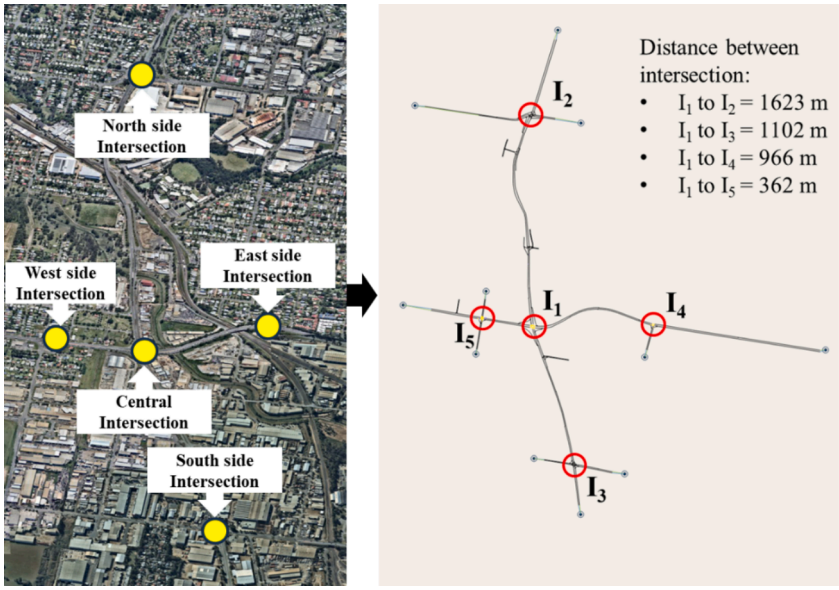


Fig. 6. Simulated traffic network and the distance between the intersections (Network centroids are marked as small blue circles with dot).

3.6. Discussion on stability of PASCAL

The proposed PASCAL signal control model is developed for realistic urban traffic networks characterised by finite link storage capacities, spillback propagation, non-linear pressure interactions, and movement-level group-based control. Under such conditions, deriving a classical Lyapunov stability proof, as commonly adopted for conventional MP control algorithms, is highly challenging.

In particular, existing analytical stability proofs for conventional MP controllers are generally established under idealised assumptions such as infinite queue storage capacity of each link, point queue representations, and linear pressure formulations. In contrast, PASCAL explicitly incorporates bounded physical storage capacity constraints of each lane and non-linear multiplicative pressure terms, which introduce state coupling and discontinuities as links approach spillback conditions.

We carefully explored the possibility of including a formal Lyapunov stability proof. However, establishing such a proof would require introducing assumptions that may not adequately reflect realistic traffic dynamics under finite link storage capacity of urban traffic networks. In particular, assumptions related to asymptotic queue growth and residual storage behaviour become difficult to justify under realistic spillback conditions. Presenting a theoretical proof based on such assumptions may therefore be misleading.

For this reason, this study focuses on a rigorous empirical assessment of stability, where the controller's stability characteristics are evaluated through extensive stress-testing simulation experiments across progressively increasing demand levels, network saturated regimes, and multiple benchmarking scenarios. The details of these analyses are presented in Section 4.

As demonstrated in the following section, across all tested scenarios, the controller consistently exhibits bounded queue evolution, delayed congestion onset, improved throughput, absence of unstable oscillatory behaviour, and robust performance under severe demand conditions. These results provide strong empirical evidence of the bounded and stable operational behaviour of the proposed control framework across a wide spectrum of traffic states.

A formal analytical stability proof for PASCAL under realistic finite-capacity spillback conditions remains an important avenue for future research.

4. Results and discussions

This section presents the evaluation of the PASCAL model against several benchmark models under varying demand and control conditions. The evaluation was performed in a simulated environment using Aimsun, the details of which are presented in Section 4.1. The detailed information on the selected benchmarking models, along with their configurations and parameters, are provided in Section 4.2.

4.1. Simulated network characteristics and demand conditions

An arterial traffic network in Brisbane, Australia (left-hand side driving) was simulated, as shown in Fig. 6. The network consists of five intersections, with the central intersection connects two major arterials (Beaudesert Rd and Riawena Rd/Granard Rd). This network, situated between Brisbane and Logan cities in Queensland, experiences heavy traffic during peak hours in all directions. The internodal distances between the intersections are also shown in Fig. 6.

The simulation was conducted for the morning peak period from 6:00 am to 9:00 am, which provides an understanding of how

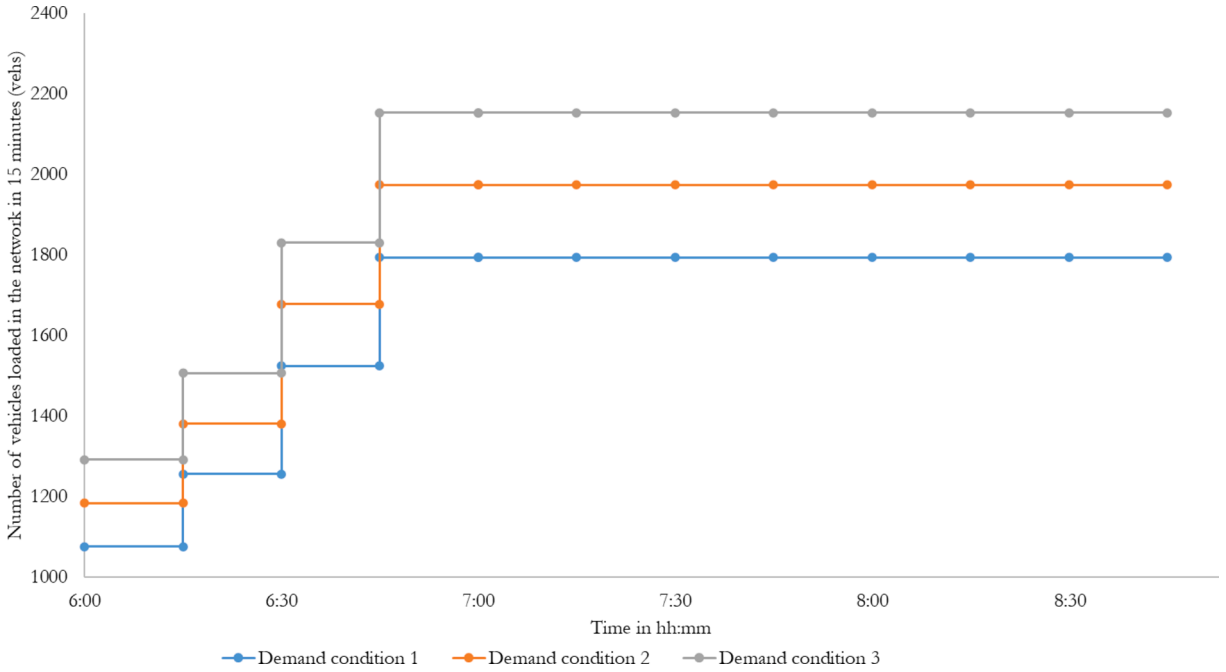


Fig. 7. Vehicle loading (every 15 min) in the Aimsun simulated network for three demand conditions (for ease of representation, uniform profiles are illustrated).

Table 1
Degree of Saturation (DoS) obtained from SIDRA for each intersection under all three demand conditions *(applicable to peak-hour demand, 7–8 am)*.

Intersection	Degree of Saturation (DoS) obtained from SIDRA		
	Demand condition 1	Demand condition 2	Demand condition 3
Central (I_1)	**0.92**	**0.98**	**1.07**
North side (I_2)	**0.88**	**0.93**	**1.02**
South side (I_3)	0.79	0.83	**0.98**
East side (I_4)	0.44	0.49	0.52
West side (I_5)	0.84	**0.89**	**0.92**

traffic congestion propagates through the network over an extended period. For each experiment, five replications of the simulation were carried out, and the average of these replications was reported for all performance measures. The headway distribution of vehicle generation in the simulation followed an exponential distribution as specified in Aimsun.

Three demand profiles (at 15-minute interval) for vehicle loading in the network were considered for the analysis, as presented in Fig. 7. Each demand profile represents the distinct congestion level over the three-hour simulation period. Specifically:

- Demand condition 1: 20,000 vehicles;
- Demand condition 2: 10 % increase in demand condition 1 (22,000 vehicles) and;
- Demand condition 3: 20 % increase in demand condition 1 (24,000 vehicles).

To ensure a realistic build-up of congestion and to avoid artefacts associated with instantaneous full network loading, the first three 15-minute intervals (6:00–6:45 am) were treated as a warm-up period. During this period, traffic demand was gradually introduced using scaling factors of 0.60, 0.70, and 0.85 relative to the respective nominal peak demand level. From 6:45 am onward, demand was maintained at the full specified level for the remainder of the simulation. A cool-down period was intentionally not included, as urban arterial performance is primarily governed by near-capacity and oversaturated conditions, while low-demand regimes can be effectively handled by conventional signal control strategies. Excluding a cool-down phase therefore allows the evaluation of PASCAL to focus explicitly on sustained congested operations.

To characterize intersection-level congestion stress under the three demand conditions, in Table 1 we present intersection-specific degree of saturation (DoS) reported by SIDRA applied to the peak-hour demand (7:00–8:00 am) for each demand condition, with cycle times optimized to minimize average delay. SIDRA is an analytical, steady state signal optimization and evaluation tool that is typically applied to a representative peak hour demand period.

Across all demand conditions, the central intersection is the most congested, as majority of the traffic passes through it. Under

Table 2
Details of the benchmarking models and their respective parameters used for evaluation.

Model	Pressure function	Control logic			Parameters
		Type	**Order**	**Cycle Length**	
M0	N/ A	Phase-based	Cyclic	Fixed cycle	SIDRA optimized timings for given demand
M1	Varaiya (2013b)		Acyclic	N/ A	Time step = 15 s
M2	Varaiya (2013b)/ Le et al. (2015)		Cyclic	Fixed cycle	• Phase configurations are same as SIDRA • Cycle lengths are same as SIDRA • Minimum green for Demand conditions 1 and 2 = 15 s • Minimum green for Demand conditions 3 = 18 s
M3	Varaiya (2013b)/ Levin et al. (2020)		Cyclic	Dynamic cycle length	• Whether a phase to be continued or move to next phase depends on the optimization process as per Levin et al. (2020) • Time step = 20 s • Maximum cycle length = 300 s
M4	Gregoire et al.- Capacity Aware		Acyclic	N/ A	Time step = 20 s
M5	PASCAL	Group-based	N/ A	N/ A	• Minimum green = 10 s • Maximum green = 70 s

Table 3
Intersection wise phase configurations and cycle lengths under all three demand conditions for model M0 and model M2.

Intersection	**Phase configurations**				**Cycle length (s)**					
					Demand condition 1		**Demand condition 2**		**Demand condition 3**	
	Phase A	**Phase B**	**Phase C**	**Phase D**	**M0**	**M2**	**M0**	**M2**	**M0**	**M2**
Central	NB Th + RT	SB Th + RT	EB Th + RT	WB Th + RT	80	80	100	100	120	100
North side	SB Th + RT	NB Th + RT	EB Th + RT	WB Th + RT	90	90	110	110	110	120
South side	NB Th + LT SB Th + LT	WB Th + RT SB LT	EB Th + RT NB LT	NB RT SB RT	80	80	90	90	90	100
East side	EB Th + WB Th	EB Th + RT	NB RT	N/ A	140	140	140	140	150	130
West side	EB Th + LT + RT	WB Th + RT SB LT	SB Th + LT + RT EB LT	NB Th + RT	100	100	110	110	140	150

demand condition 2, central, north side, and west side intersections are operating close to capacity (exhibit higher DoS). Under demand condition 3, congestion further propagates to the south-side intersection, indicating network-wide saturation effects as demand increases.

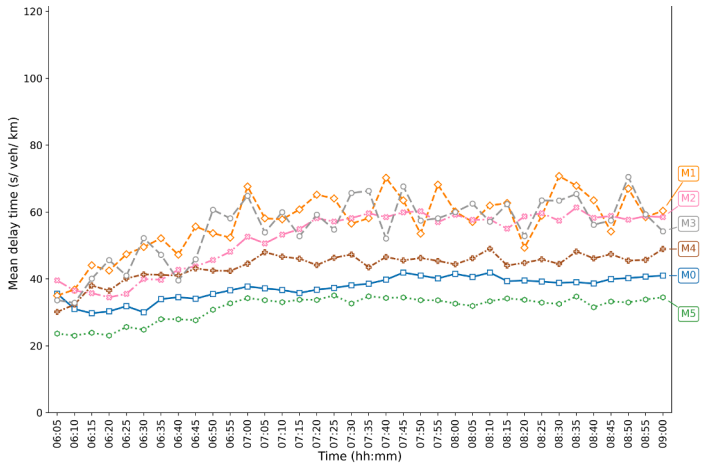
4.2. Benchmarking models

Since existing MP control techniques are broadly classified into acyclic and cyclic, one representative model from each group was selected for benchmarking. This includes the seminal MP control proposed by Varaiya (2013b) from the acyclic group, and the MP control by Le et al. (2015) from the cyclic group. As PASCAL incorporates storage capacity into the formulation of its pressure function, it is necessary to evaluate its efficiency against an MP control model that also accounts for storage capacity. For this purpose, the capacity aware backpressure control by Gregoire et al. (2014) was selected. Furthermore, PASCAL was benchmarked against the MP control proposed by Levin et al. (2020), which maintains a cyclic structure while implementing a dynamic cycle length. Additionally, PASCAL was also compared with the SIDRA-optimized signal timings.

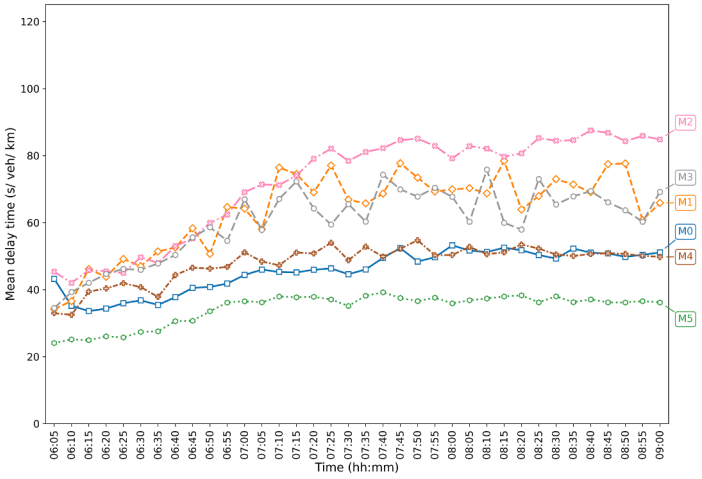
The performance of each model depends on several parameters. For example, in acyclic MP control models such as the MP control by Varaiya (2013b) and the capacity aware MP by Gregoire et al. (2014), the duration of each time step significantly affects the signal control performance. In contrast, the performance of cyclic MP models such as the MP control by Le et al. (2015) depends on phase configurations, cycle time, and minimum green time. Similarly, the MP control proposed by Levin et al. (2020) relies on the duration of the time step, phase configurations, and the maximum cycle time at each intersection. In this study, the parameters for each model were set to the values that performed optimally under the tested demand and control conditions. The codes along with the selected model parameters are available and can be reproduced for further testing. The details are provided in Section 4.7. Furthermore, all models were allotted equal amber and clearance red times of 4 s to ensure a fair comparison.

It is also important to note that both the MP control by Le et al. (2015) and the MP control by Levin et al. (2020) employ the same weight and pressure functions defined by Varaiya (2013b), although their control logics differ from Varaiya's (2013b) formulation. Table 2 summarizes the benchmarking models used for evaluation along with their respective parameter settings. For ease of discussion, we number the models from M0 to M5. Table 3 shows the phase configurations and cycle lengths for each intersection for fixed cycle length models (M0 and M2).

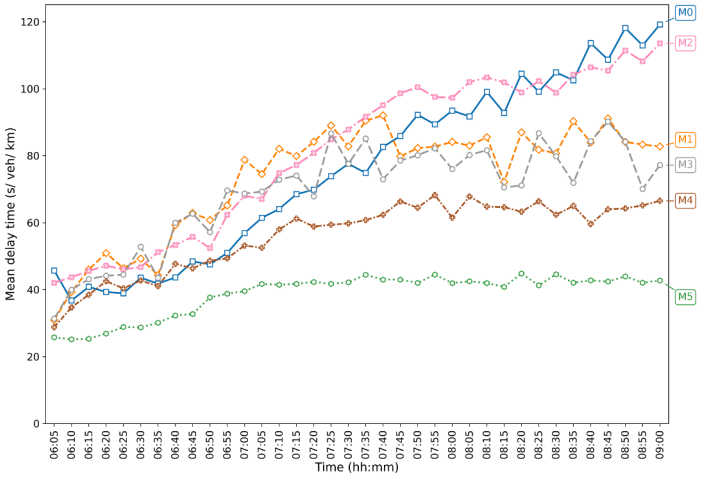
In the existing MP control studies, the pressure function of a phase is calculated as the sum of the weights multiplied by the respective saturation flows of the movements within that phase. Since saturation flows for through movements are typically higher



(a) Demand condition 1



(b) Demand condition 2



(c) Demand condition 3

Fig. 8. Time series of mean (over 5 min) delay time for six signal control models (Model M0 to M5) under three demand conditions. For ease of visual comparison, the axis scales are same for each of the demand conditions.

than for turning movements, the acyclic MP model tends to prioritize the through movements over the right turning movements, especially under high-demand conditions. This imbalance can result in increased delays for turning movements. We tested the performance of all the benchmarking models, including PASCAL, both with and without considering saturation flow. The results showed that all the models performed better without considering the saturation flow rates of the movements. Therefore, the MP control models used for benchmarking in this study exclude the multiplication by saturation flow rates when computing pressure.

Another important consideration concerns the calculation of pressure for movements operating under shared lane configurations. In the current PASCAL formulation, for lanes with shared movements the signal controller does not know which specific movement each vehicle will take. In this case, two approaches can be adopted to estimate the pressure of the movements associated with the shared lane.

In the first approach (proportional splitting using turning ratios), average historical turning proportions for the shared movements can be used to split both the number of vehicles in the shared lane and the available storage capacity. The drawback of this method is that it does not fully capture the stochastic variation in turning proportions, which is common in real-world conditions. In addition, this splitting may underestimate the demand and usable storage capacity of the movements. For example, if the leftmost lane is shared between through and left-turn movements, there may be situations where most vehicles in that lane are turning left. In such cases, applying fixed turning proportions can underestimate the pressure of the left-turn movement. Ideally, it is better to slightly overestimate the pressure and serve a movement than to underestimate the pressure and fail to serve the movement which may further alleviate the spillback.

In the second approach (conservative allocation based on full shared-lane occupancy), the pressure calculated using the total number of vehicles and the total storage capacity of the shared lane is assigned to all movements associated with that lane. This may slightly overestimate the actual pressure of each shared movement and can sometimes lead to over-prioritisation, but it avoids the need to explicitly model the stochastic nature of turning behaviour.

In the presented results, the second approach has been adopted for shared lanes. The total number of vehicles and the total storage capacity of the shared lane are used to calculate the pressure for each movement associated with that lane. This implementation choice avoids reliance on uncertain turning-proportion estimates and is consistent with the observable data typically available in max-pressure control systems.

4.3. Evaluation

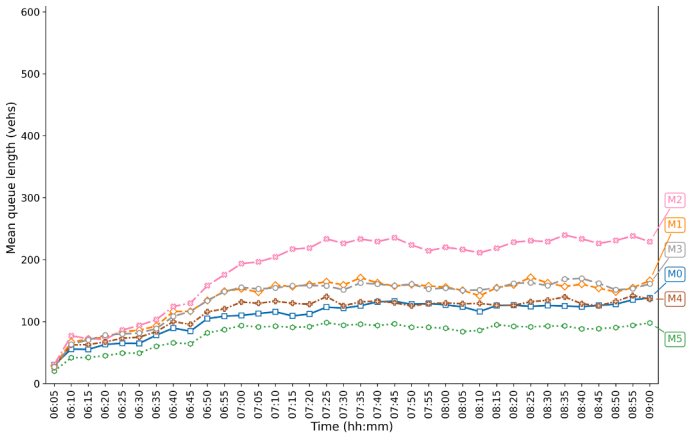
Numerical experiments were conducted to evaluate the effectiveness of the proposed control strategy using various network-level performance metrics. Mean delay time, mean queue length, and the number of vehicles in the network were computed at 5-minute intervals over the 3-hour simulation period, while aggregated mean speed was computed over the entire simulation horizon. The performance measures are defined in Aimsun as follows:

- **Mean delay time (s/ veh/ km):** For a given time interval (5 min in this study), Aimsun considers the vehicles that exited the network during that interval. For each vehicle, it computes the delay accumulated in each section. It then divides the accumulated delay by the distance travelled by the corresponding vehicle and averages these values over all vehicles that exited the network during that interval.
- **Mean queue length (veh):** This is the average queue length observed across all links in the network within a specific time period, expressed in vehicles.
- **Number of vehicles in the network (veh):** This is the total number of vehicles present in the network at a time instant. For a given demand, it is inversely proportional to the trip completions, where higher number of vehicles in successive intervals indicate fewer trip completions across the network, and therefore, lower vehicle throughput at intersections.
- **Aggregated space mean speed (km/ hr):** It is computed as the ratio of the total distance travelled by all vehicles to the total travel time taken during the entire simulation period.

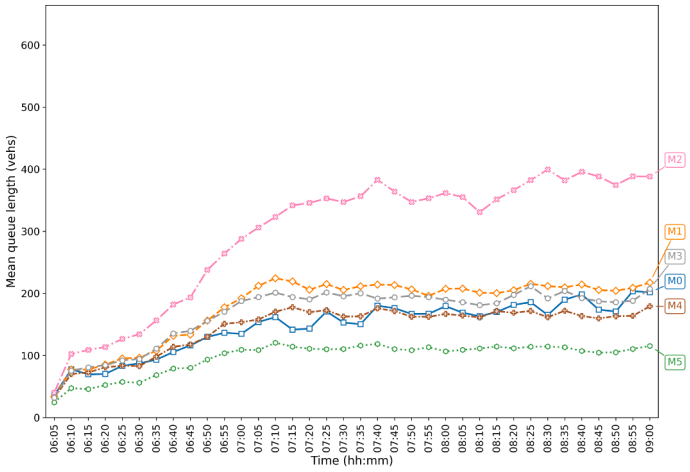
Fig. 8 illustrates the time series of the mean delay time for all models (M0 to M5, refer to Table 2) under the three demand conditions. Across all cases, model M5 (PASCAL) consistently achieves the lowest mean delay, with performance gains becoming more pronounced as demand increases.

This behaviour can be attributed to two key features of PASCAL. First, the group-based control strategy enables the simultaneous activation of non-conflicting high-pressure movements, thereby increasing effective discharge rates under unbalanced demand. Second, the pressure formulation explicitly accounts for vehicle accumulation in the candidate lane(s) and storage availability of both the candidate lane(s) and the downstream link, allowing green time to be allocated where it can be used most effectively. Together, these mechanisms reduce wasted green time and ensure adequate service for critical movements, resulting in substantially lower delays compared with the benchmark models. In addition, model M5 exhibits significantly lower temporal variations in delay throughout the simulation. This indicates that travel times remain stable even if network loading increases, demonstrating the ability of PASCAL to adapt to changing traffic conditions and delay the onset of congestion through its group-based control, pressure function, CMSP mechanism, and dynamic green allocation.

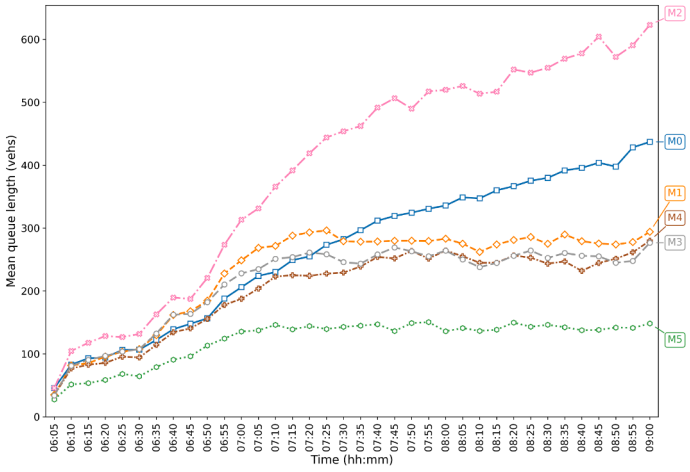
Model M0 (fixed-time control optimized by SIDRA) performs well under demand condition 1 but deteriorates as demand increases. Since signal timings are optimized for peak-hour conditions, they cannot adequately respond to spatiotemporal variations in demand, particularly under heavily congested and unbalanced conditions. This limitation becomes evident under demand condition 3, where delays increase substantially.



(a) Demand condition 1



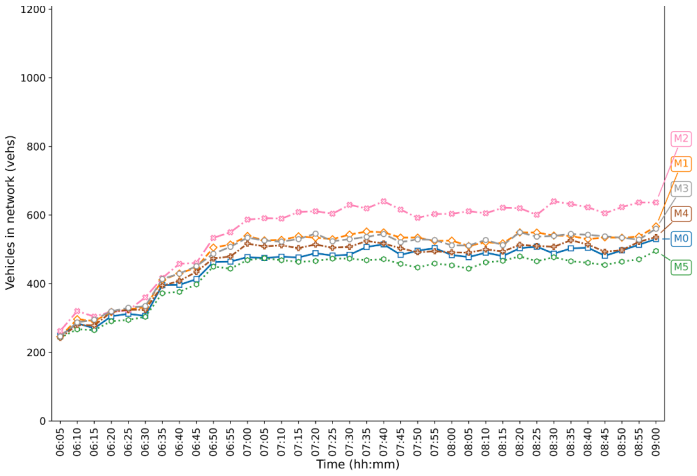
(b) Demand condition 2



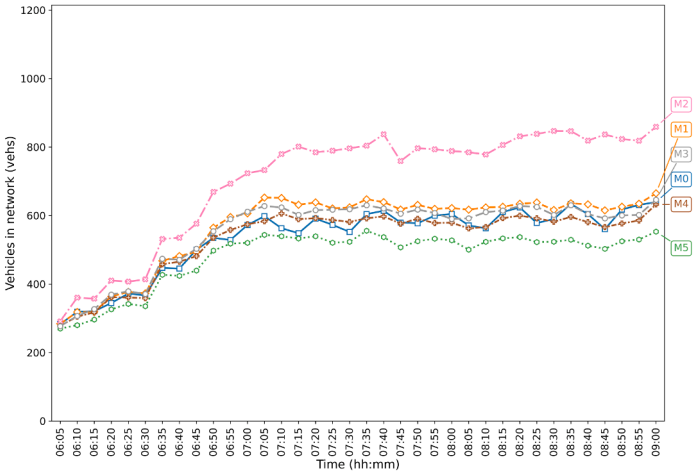
(c) Demand condition 3

Fig. 9. Mean queue length observed in the network resulted by six signal control models (Model M0 to M5) under demand conditions 1, 2, and 3.

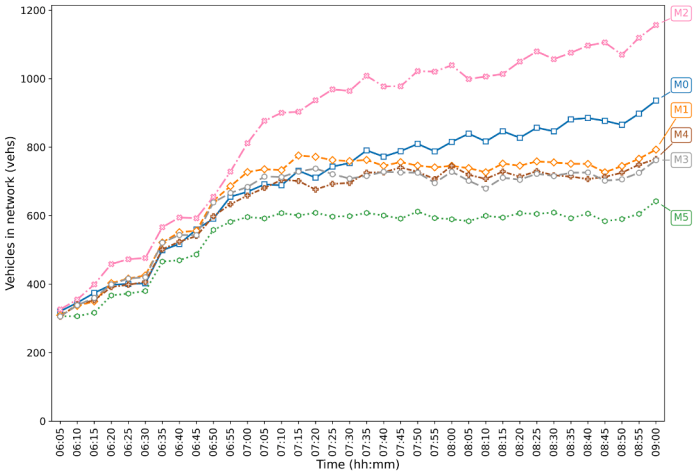
Model M1 (Varaiya, 2013b) exhibits oscillatory delay patterns, particularly in the later stage of the simulation. As an acyclic controller, it repeatedly prioritises high-pressure phases, allowing lower-demand phases to accumulate vehicles over time. When these phases are eventually served, the delay temporarily decreases before increasing again. This pattern is most evident at intersections



(a) Demand condition 1



(b) Demand condition 2



(c) Demand condition 3

Fig. 10. Number of vehicles in the network observed in the network resulted by six signal control models (Model M0 to M5) under demand conditions 1, 2, and 3.

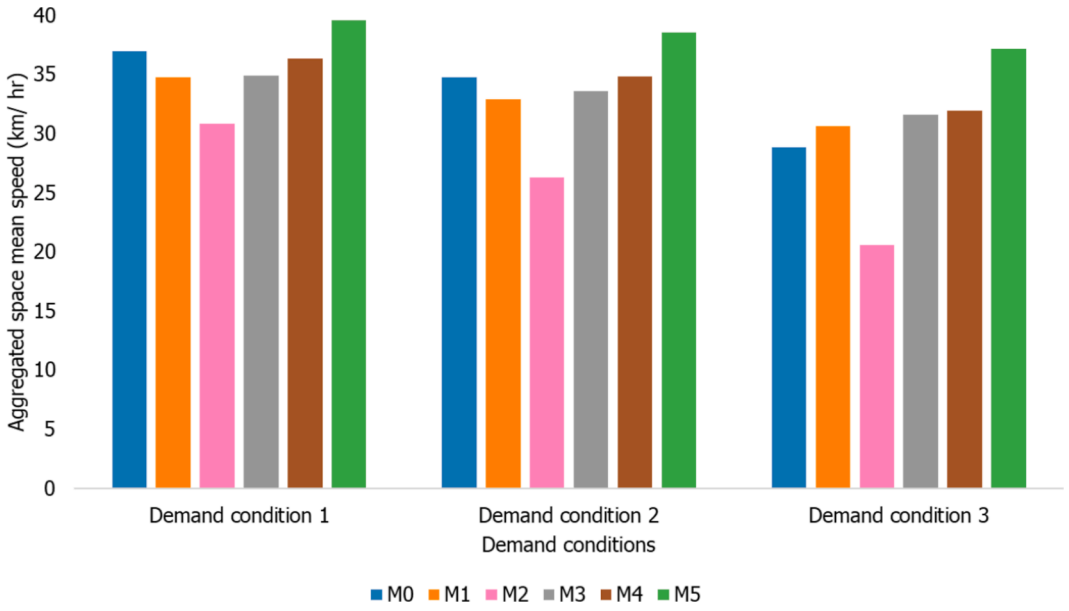


Fig. 11. Aggregated space mean speed by six signal control models (Model M0 to M5) under demand conditions 1, 2, and 3.

where some approaches experience significantly higher demand than others.

Model M2 (Le et al., 2015) performs slightly better than model M1 under low demand but deteriorates significantly as demand increases. Although it uses the same phase structure as model M0, green time is allocated using an exponential weighting of phase pressure. This disproportionately favours high-pressure phases, leaving low-pressure phases underserved. As congestion increases, this imbalance leads to substantial delay accumulation.

Model M3 (Levin et al., 2020) shows performance similar to model M1, with marginal improvement across all demand conditions. While both models use similar pressure formulations, model M3 incorporates a dynamic cycle length, enabling more adaptive green allocation and slightly improved delay performance.

Model M4 (capacity-aware MP, Gregoire et al., 2014) performs better than the other signal control models, except for model M5, particularly under high and unstable demand. This can be attributed to its consideration of storage availability in the pressure function, similar to PASCAL (Model M5). However, since it is an acyclic model, it cannot achieve the same level of performance as model M5′s group-based logic.

Furthermore, there are structural differences between the pressure functions of models M4 and M5, even though both consider storage capacities. This necessitates further benchmarking of the two models under the same group-based control logic to explicitly differentiate the effectiveness of their pressure functions.

Fig. 9 shows the time series of the mean queue length for all the signal control models under the three demand conditions. Consistent with the mean delay time results, model M5 (PASCAL) shows the lowest mean queue length over the simulation period under all three demand conditions. The group-based control logic activates higher-pressure movements not only as the critical movement but also as complementary movements, thereby preventing larger queue build-ups on links compared with the other models.

The overall performance of models M0 and M4 was comparable to their respective delay results under all three demand conditions. Nevertheless, model M4 results in much lower queue build-ups under high and unstable demand conditions than model M0 due to its consideration of storage capacity in the pressure function. In contrast, model M0′s fixed phase sequence, optimized for peak-hour demand, results in much higher queue lengths under such high and unstable demand conditions.

Model M1 performs nearly the same as model M3 under the low-demand condition, as both models use the same weight and pressure function, resulting in similar phase activations under stable and low vehicle loading. However, as demand increases, the dynamic cycle length structure of model M3 allocates more green time to the critical movements than model M1, resulting in much lower queue build-ups for model M3 than for model M1. Furthermore, under the high and unstable demand condition (after 7:30 am in demand condition 3, model M3, with its dynamic cycle length and fixed phase sequence, could discharge as many vehicles as model M4, which considers storage capacity.

Model M2 produces much higher queue lengths than the other models, particularly under demand conditions 2 and 3. Since the exponential formulation of model M2 allocates the majority of the cycle time to phases with comparatively higher pressure, phases with lower pressure often receive only minimum green time. However, a minimum green time may not be sufficient to discharge all the vehicles for a comparatively lower-pressure movement, leading to split failures under higher demand.

Fig. 10 illustrates the time series of the number of vehicles in the network produced by different signal control models, which essentially indicates the overall network throughput achieved by the models. When network throughput decreases, the number of

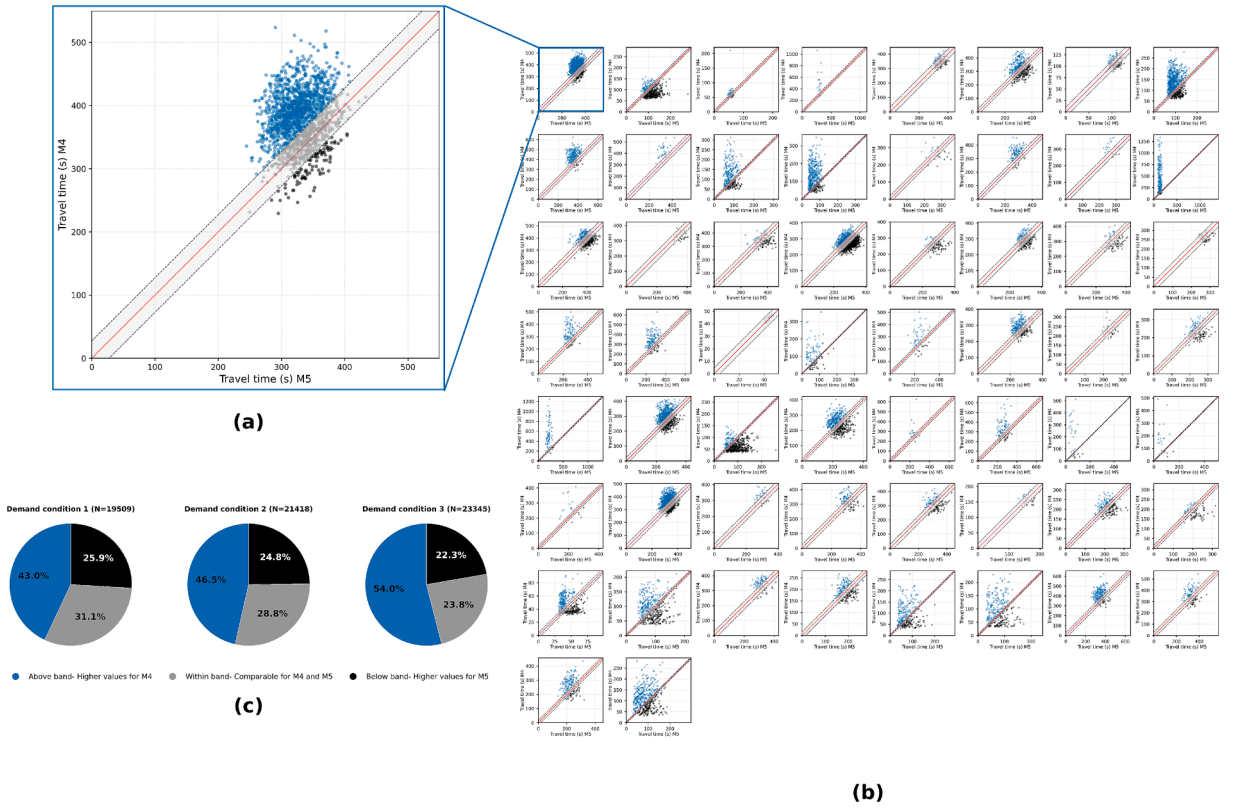


Fig. 12. In a clockwise direction starting from top left figure: (a) Scatterplot of per-vehicle travel time resulted by M4 (y-axis values) and M5 (x-axis values) for a particular OD pair under demand condition 1 where the blue, grey and black points represent higher travel times under M4, comparable travel times between the models, and higher travel times under M5, respectively; (b) Pairwise scatterplots of per-vehicle travel times with M4 and M5 for all OD pairs in the network; and (c) Percentages of trips classified as blue, grey, and black across all three demand conditions.

vehicles in the network increases, and vice versa.

Model M5 (PASCAL) consistently results in the lowest number of vehicles in the network over time, particularly under demand conditions 2 and 3, indicating higher intersection throughput and, correspondingly, higher trip completion. As mentioned earlier, this performance is attributed to the pressure function formulation and the group-based logic, which result in a higher number of activations and longer green times for the critical movements. model M5 also exhibits the most delayed onset of congestion among all models. This is examined further under higher demand levels in Section 4.4.

Under low demand condition, the performance of all models is not significantly different, as the signal control models tend to exhibit similar throughput. However, as demand increases, models M1, M3, and M4 show comparable performance, with model M3 performing slightly better in the later part of the simulation under demand condition 3. The dynamic cycle length contributes to model M3′s performance under congested scenarios.

Consistent with the mean queue length results, model M0 reports a much higher number of vehicles in the network under high demand, except for model M2 which shows the lowest throughput across the network. Furthermore, under high and unstable demand conditions, models M0 and M2 show a much earlier onset of congestion than the other models due to the formation of residual queues at high demand levels, leading to a larger number of vehicles remaining in the network over time.

Fig. 11 presents the aggregated space mean speed over the 3-hour simulation period, and model M5 consistently achieves higher aggregated values across all three demand conditions, indicating lower congestion, smoother flow, and shorter stop times at intersections. Since model M5 can adapt better to changing traffic conditions than the other benchmark models, the difference in space mean speed between model M5 and the other models increases as demand rises. The activation of high-pressure movements as complementary movements in model M5 helps reduce vehicle stop times, thereby reducing trip completion times and increasing space mean speed. Furthermore, under low demand, the signal control models tend to perform similarly.

Model M0 reports higher space mean speed under low- to moderate-demand conditions due to the optimized performance of fixed-time control under stable demand. However, under congested and near-saturated conditions, model M0 cannot accommodate the varying demands at the conflicting approaches and therefore records much lower space mean speed than under lower demand conditions.

Models M1 and M3 perform almost similarly across the demand conditions. Nevertheless, model M3 reports slightly higher space mean speed under higher demand due to its adoption of dynamic cycle length.

Table 4
Details of the benchmarking models and their respective parameters used for group-based control evaluation.

Model	Pressure function	Control logic	Parameters
M5	PASCAL	Group-based	• Minimum green = 10 s • Maximum green = 70 s
M6	Varaiya (2013b)		
M7	Gregoire et al. (2014) – Capacity Aware		

Model M2 experiences persistent split failures, leading to repeated vehicle stops at signals. This leads to longer trip completion times and, consequently, the lowest space mean speed among all models across all three demand conditions.

Model M4 shows aggregated space mean speeds comparable to model M0 under demand conditions 1 and 2. However, under demand condition 3, model M4 records the second-highest space mean speed due to its consideration of storage capacity, as mentioned earlier.

In Fig. 12 we present OD pairwise travel time scatter plots between models M4 and M5. Since both of these models use storage capacities in their pressure functions, the aim of this experiment was to compare how each trip experienced the travel time between the same OD pair under the two models.

Fig. 12(a) first shows an example scatterplot for one OD pair under demand condition 1. The x-axis gives the travel time (in seconds) for each trip under M5, and the y-axis gives the travel time for the same trip under M4. Travel time can be computed as the difference between a vehicle's entry and exit times in the network. The red dotted diagonal line represents equal travel times for both models ($y = x$). Around this line, a band is drawn and the half-width of the band is 10 % of the free-flow travel time for that OD pair. Points within this band (grey) are considered to have comparable travel times for the two models and attributed to the stochastic behaviour of the signal control models. Points above the band (blue) indicate trips where M4 produces higher travel times than M5, while points below the band (black) indicate trips where M5 performs worse than M4.

Using the same convention, Fig. 12(b) shows the scatterplots for all OD pairs in the network under demand condition 1. For most OD pairs, especially where the data cloud is large, the proportion of blue points is much higher than that of black points, indicating that users experience longer travel times under M4 than under M5. This is consistent with the group-based nature of M5, where movements with higher demand are often served repeatedly as complementary movements, resulting in lower travel times for major OD pairs. In several OD pairs the point cloud is tall and relatively narrow, suggesting that M5 yields travel times in a narrow range, whereas M4 produces both higher and more variable travel times. There are a few OD pairs where the proportions of black points are higher, showing higher travel times under M5. Furthermore, a small number of OD pairs show an equivalent proportion of dispersed data points (both blue and black) on both sides of the band.

Fig. 12(c) summarises the patterns of the scatter plots across demand conditions. As demand increases, the percentage of blue points increases, and in contrast, the percentages of grey and black points reduce. Although travel times increase for both models under demand conditions 2 and 3 as compared to demand condition 1, the increase is more pronounced for M4. This resulted many points to move above the band, increasing the proportion of blue points in high demand conditions. This also suggests that, under higher congestion, M5 can achieve higher throughput and substantially lower travel times than M4.

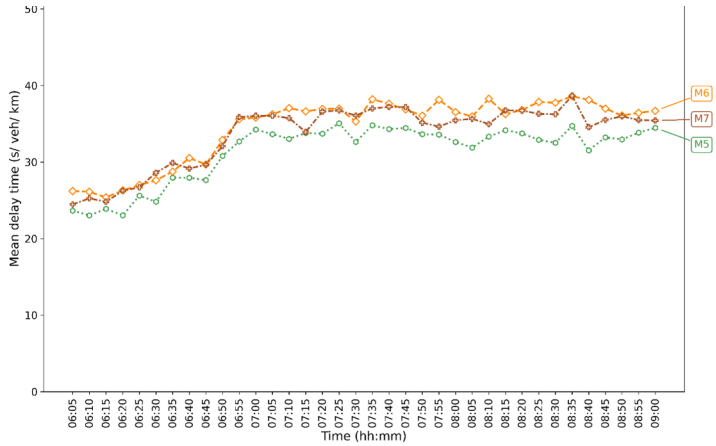
Overall, these results show that even though both models include storage capacity in their pressure formulations, the group-based control logic and the fairness mechanism based on the critical movement selection pool (CMSP) in model M5 strongly influenced performance. Therefore, it is important to compare the performance of these two models where the group-based control logic and CMSP are applied to the capacity-aware MP model (M4) to assess the benefits of the PASCAL pressure function (M5).

4.4. Evaluation of group-based control

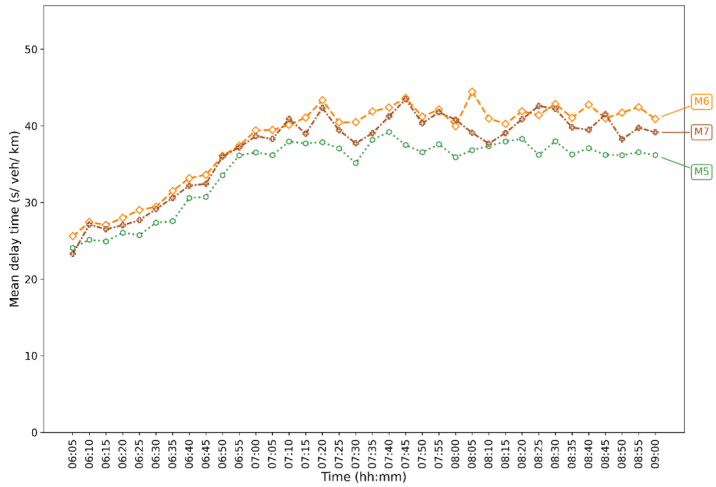
From the results of the previous section, it was observed that the performances of models M1, M3 and M4 were better than those of the other models, except for PASCAL (M5). Hence, it is important to evaluate how their pressure functions perform within the same group-based control framework. Among these three models, M1 and M3 use the same pressure function, but their control logics differ. Since the control logic in this experiment is same for the models, i.e., group-based control, effectively M1 and M3 would perform the same. In contrast, M4 employs a different pressure function from M1. Therefore, in this section, the pressure functions of M1 (Varaiya, 2013b) and M4 (Gregoire et al., 2014) were applied in conjunction with the group-based control framework adopted by PASCAL. For ease of discussion, these models are numbered as M6 and M7, respectively. Table 4 summarises the models and the parameters used for this group-based control evaluation.

Similar to the previous section, (a) mean delay time, (b) mean queue length, and (c) number of vehicles in the network were analysed for models M5, M6, and M7 under the three demand conditions. Fig. 13 shows the mean delay times obtained from the models, and their performances are comparable under all three demand conditions. This is expected, as the flexibility of group-based control in activating various combinations of movements facilitates the repeated serving of critical movements. Nevertheless, model M5 (PASCAL) still achieves lower mean delay times than the other two models, particularly under high and unstable demand. The better performance of model M5 can be linked to the way its pressure function balances demand growth with storage availability, which is further facilitated by repeated activations of critical movements based on the group-based control logic. Therefore, model M5 performs better than the other two models, even though all three models implement group-based control logic. This is observed more clearly in the next section under higher demand levels.

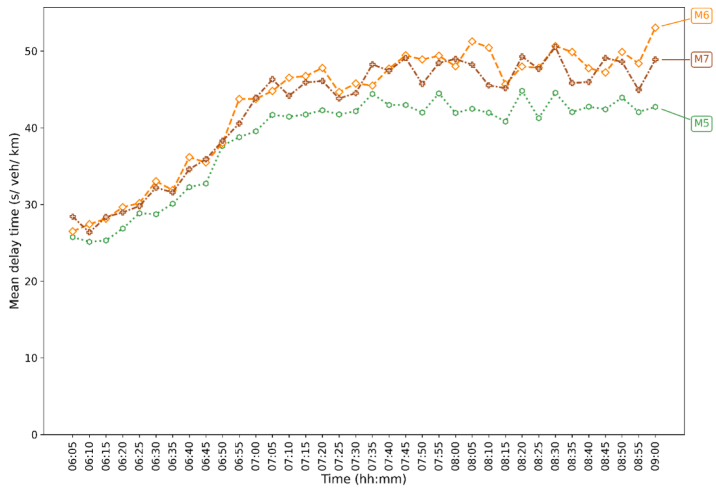
Models M6 (Varaiya, 2013b pressure function) and M7 (Gregoire et al., 2014 pressure function) show similar performance across



(a) Demand condition 1



(b) Demand condition 2



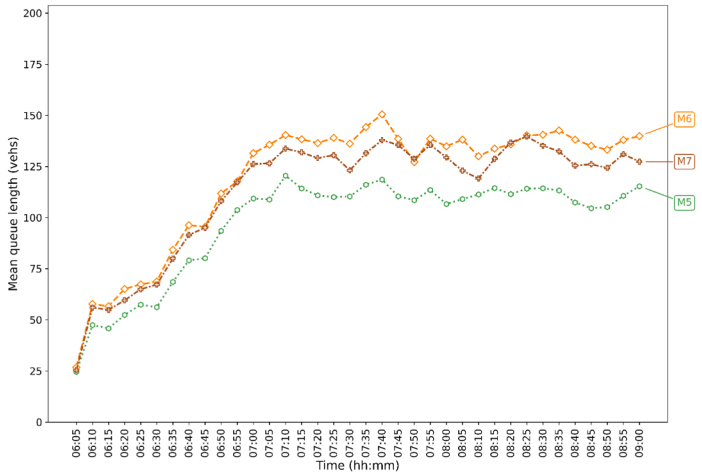
(c) Demand condition 3

Fig. 13. Mean delay time observed in the network resulted by models M5, M6, and M7 under demand conditions 1, 2, and 3.

all three demand conditions. Nevertheless, model M7 performs slightly better than model M6 across all three demand conditions due to model M7′s storage availability consideration in its pressure function. Note that although models M5 and M7 both consider storage availability in their pressure functions, their functional forms differ. Specifically, in model M5, the storage-availability terms act as



(a) Demand condition 1



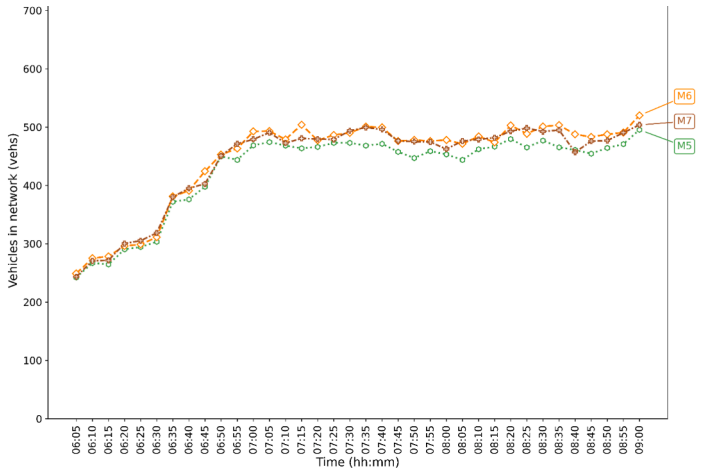
(b) Demand condition 2



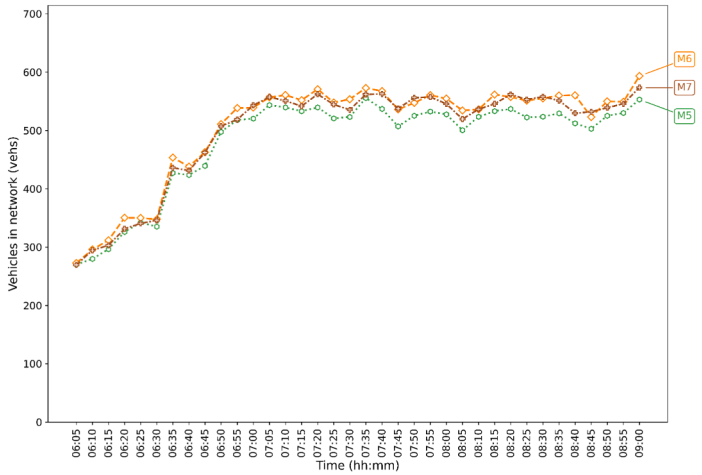
(c) Demand condition 3

Fig. 14. Mean queue length in the network resulted by models M5, M6, and M7 under demand conditions 1, 2, and 3.

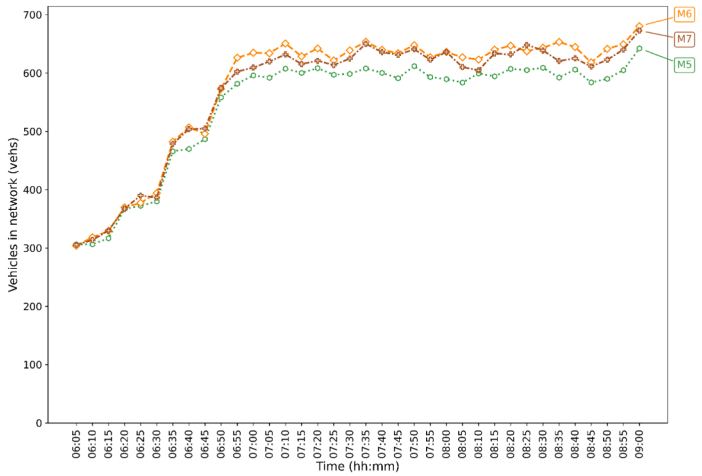
regulatory factors on the growing demand, whereas in model M7, the candidate lane(s) storage availability is penalised by the downstream storage availability, and demand is not considered directly. This structural difference leads to better performance for model M5 than for model M7, which is further demonstrated in the next section.



(a) Demand condition 1



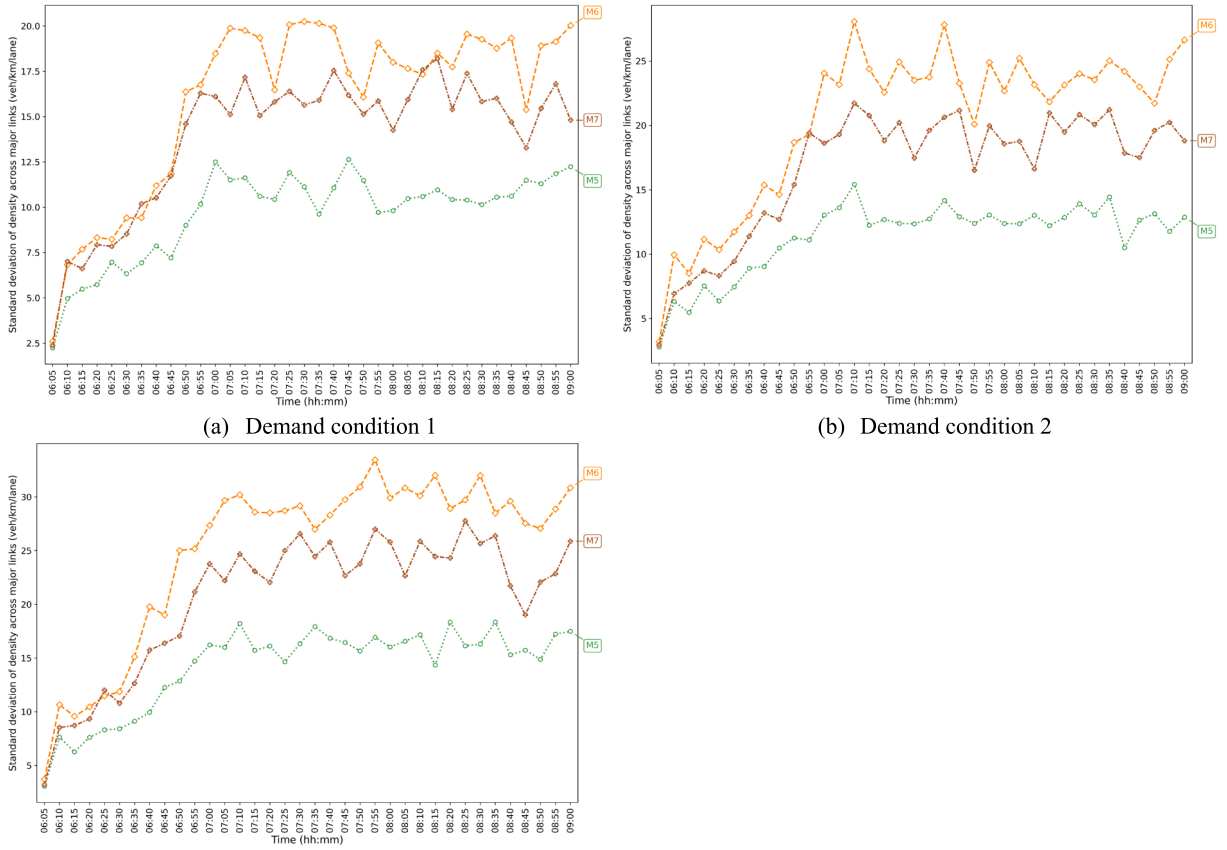
(b) Demand condition 2



(c) Demand condition 3

Fig. 15. Vehicles in the network under group-based signal control models.

Fig. 14 shows the mean queue length obtained from the three group-based models. The effect of the pressure function in model M5 is more evident in the mean queue length trends across all demand conditions. model M7 performs better than model M6 in dissipating queues, due to the storage availability factor in its pressure function. In contrast, the pressure function of model M6 only computes the



(a) Demand condition 1

(b) Demand condition 2

(c) Demand condition 3

Fig. 16. Standard deviation of density across major links with model M5, M6, and M7 under demand conditions 1, 2, and 3.

difference in the number of vehicles between the candidate lane(s) and the downstream link, which may yield a higher pressure for a movement when the downstream link is short. However, activating such movements may waste green time if the downstream link cannot accommodate the discharged vehicles. This once again highlights the importance of pressure function formulation in addition to the group-based control logic.

Fig. 15 illustrates the number of vehicles in the network under the three group-based models. Although the variations are similar across all three models, model M5 consistently maintains a lower number of vehicles in the network than the other two models.

Under low and stable demand conditions, all three models perform comparatively similar to each other. However, under high and unstable demand scenarios, the advantage of the PASCAL pressure function over the other two models becomes evident. Clearly, model M6, which uses Varaiya's pressure function, is limited because it does not consider storage availability. Nonetheless, the pressure function of model M7 (Gregoire et al., 2014) can be simplified as the ratio of the number of vehicles to the storage capacities of the candidate lane(s) and the downstream link, without explicit consideration of the growing demand in the candidate lane(s). Therefore, the advantage of the PASCAL pressure function (M5) over the capacity-aware pressure function (M7) can be understood more clearly under more congested regimes. Accordingly, in the next section, we further increase the demand and test the performance of these three models under more congested traffic conditions. Before that, we examine another important performance metric, spatial density heterogeneity across the network.

MP-based controls can reduce density heterogeneity across network links, which in turn improves network throughput and effective capacity utilisation, as reported by Tsitsokas et al. (2023), and Su et al. (2023). Hence, we have conducted an additional analysis of network density heterogeneity, following the approach presented by Su et al. (2023). Specifically, the temporal variation in the standard deviation of density is evaluated across the major links in the network under demand conditions 1, 2, and 3, by comparing PASCAL (M5) with the other group-based signal control models (M6 and M7) as shown in Fig. 16.

It can be observed from Fig. 16 that the standard deviation of density is consistently lower for the model M5 compared with models M6 and M7, with the differences becoming more prominent under high demand conditions and during later stages of simulation when network becomes more congested. This indicates PASCAL is more effective in maintaining spatial homogeneity of densities across the network.

This reduction in density is directly linked to improved network capacity utilisation. A more homogeneous distribution of vehicles reduces the likelihood of localised congestion and spillback, thereby improving the effective throughput of the network. In PASCAL,

Table 5
Degree of Saturation (DoS) obtained from SIDRA for each intersection under demand conditions 4 and 5

Intersection	Degree of Saturation (DoS) obtained from SIDRA (phase based)	
	Demand condition 4 (30 % more to Demand condition 1	Demand condition 5 (40 % more to Demand condition 1
Central	1.15	1.24
North side	1.10	1.19
South side	1.06	1.15
East side	0.55	0.58
West side	0.94	0.99

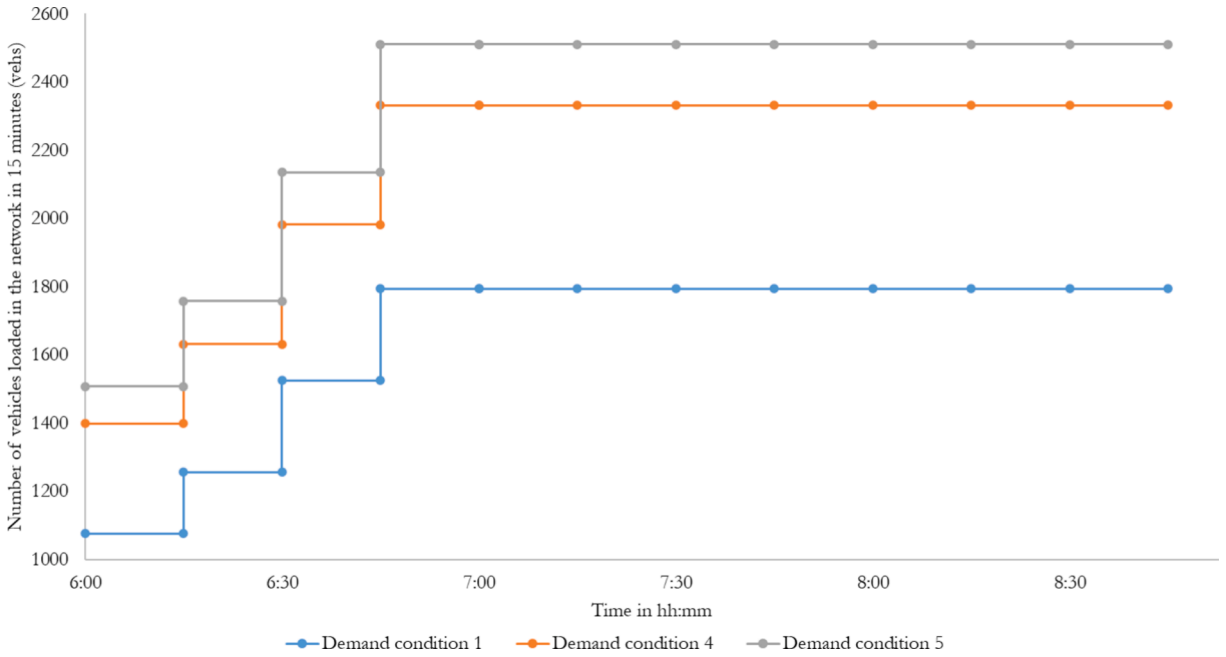
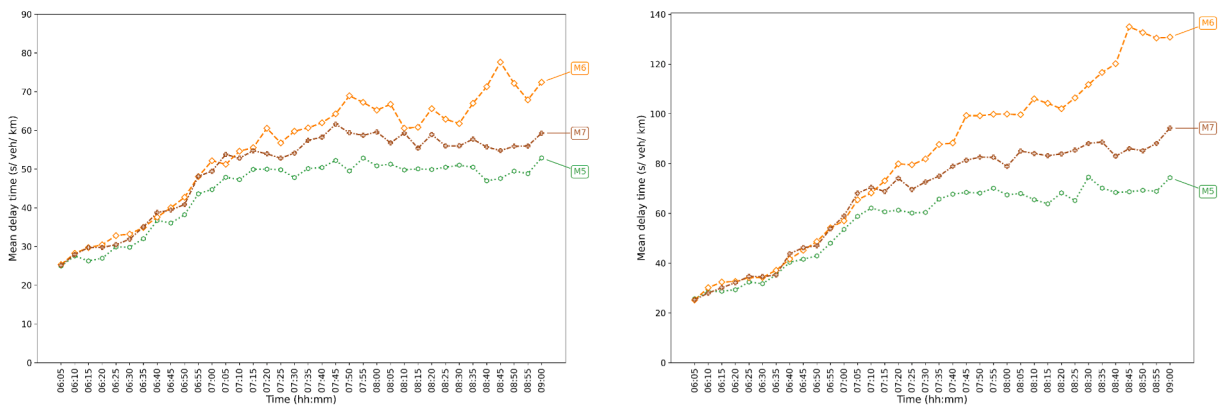
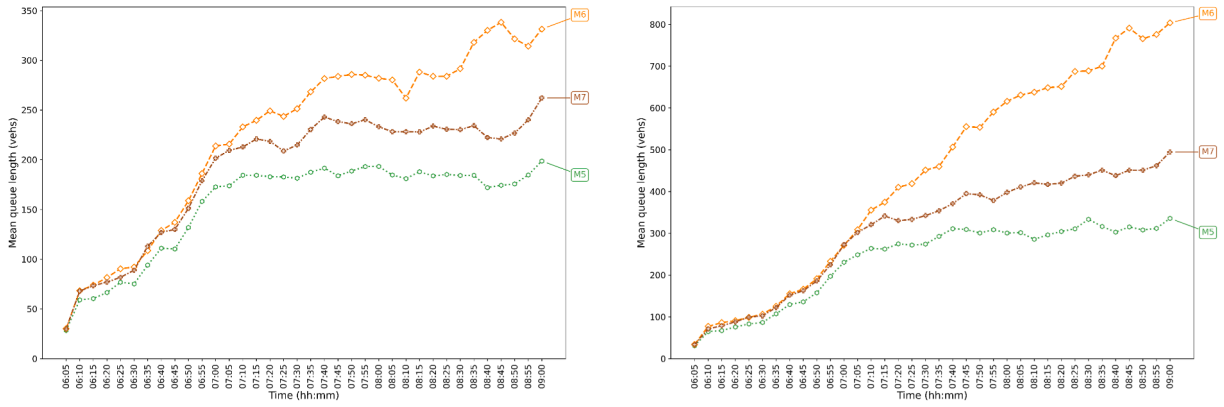


Fig. 17. Vehicle loading (every 15 min) in the Aimsun simulated network for demand conditions 4 and 5 (Demand condition 1 is shown as a reference to explain the increase in vehicle loading and for ease of representation, uniform profiles are illustrated).



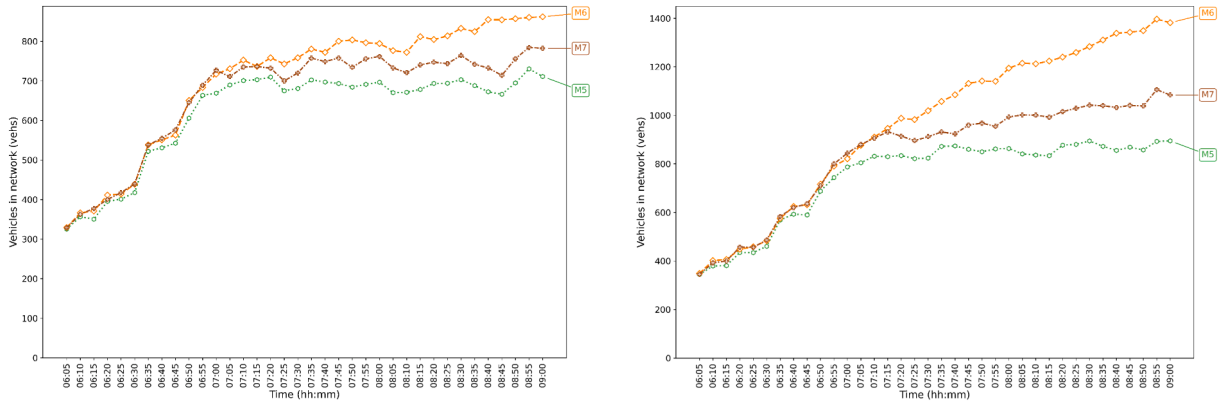
(a) Demand condition 4 (b) Demand condition 5

Fig. 18. Mean delay time under demand conditions 4 and 5 resulted by models M5, M6, and M7.



(a) Demand condition 4 (b) Demand condition 5

Fig. 19. Mean queue length under demand conditions 4 and 5 resulted by models M5, M6, and M7.



Demand conditions 4 and 5

Fig. 20. Vehicles in the network under demand conditions 4 and 5 resulted by models M5, M6, and M7.

this is achieved by the pressure formulation, which explicitly accounts for both upstream and downstream storage constraints. As a result, movements approaching spillback conditions are prioritized, while movements with insufficient downstream capacity are supressed, leading to a balanced distribution of traffic across the network.

Consequently, the improved density homogeneity observed under PASCAL contributes to enhanced network throughput and more efficient utilisation of network capacity, which is also consistent with the lower delays (Fig. 13), lower queue length (Fig. 14) and reduced number of vehicles in the network observed in the earlier results (Fig. 15(c)).

4.5. Evaluation of group-based models under further higher demand conditions

For this evaluation, two additional demand conditions were analysed, demand condition 4 and demand condition 5. In these conditions, the vehicle demand of condition 1 was increased by 30 % and 40 %, respectively. The total number of vehicles generated over the 3-hour simulation period was 26,000 under demand condition 4 and 28,000 under demand condition 5. Table 5 presents the degree of saturation (DoS) for the intersections (under phased based) in the network under these two demand conditions, as obtained from SIDRA.

Fig. 17 shows the number of vehicles loaded into the network in every 15 min time interval under demand conditions 4 and 5 considered in the evaluation. For comparison, demand condition 1 is also included in the figure to highlight how the demand profiles of conditions 4 and 5 differ from the base case.

Fig. 18 shows the mean delay times under demand conditions 4 and 5 for models M5, M6, and M7. These demand conditions clearly demonstrate the superiority of the PASCAL pressure function (Model M5) compared to the other two pressure functions. Model M5 yields much lower mean delay times under both demand conditions. In this evaluation, the network is heavily congested, resulting in reduced storage capacities in both candidate lane(s) and downstream link for the movements at most intersections. The distinctive performance gap between models M5 and M6 reflects the inclusion of storage constraints in PASCAL pressure function, which helps avoid serving spilled over downstream links and gives priority to overflown right-turn lanes when needed.

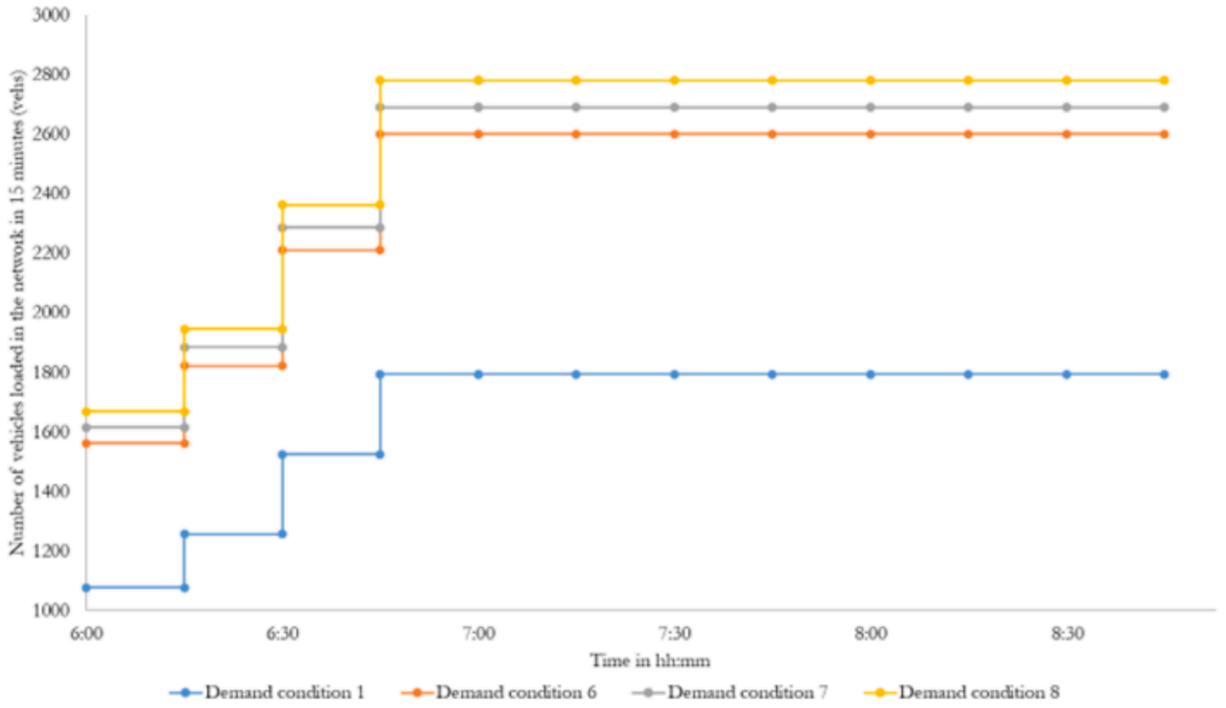


Fig. 21. Vehicle loading (every 15 min) in the Aimsun simulated network for demand conditions 6, 7, and 8 (Demand condition 1 is shown as a reference to explain the increase in vehicle loading and for ease of representation, uniform profiles are illustrated).

Even though model M7 achieves much lower mean delays than model M6, it still produces much higher mean delays than model M5. The superior performance of model M5 over model M7 is associated with the structural difference between their pressure functions, which can be understood through a simple example. Consider a scenario with two conflicting movements, A and B. Suppose both have the same ratio of vehicles to storage capacity in their respective candidate lane(s), which is possible because storage capacity depends on link length. If movement A has significantly higher demand than movement B, and both have sufficient downstream storage capacity, model M7 would assign the same pressure to both. In contrast, model M5 would prioritise movement A because of its higher demand. Consequently, model M5 performs significantly better than model M7 under demand conditions 4 and 5.

Fig. 19 shows the mean queue length under demand conditions 4 and 5 for the three models. Consistent with the previous results, model M5 produces much lower mean queue lengths than both model M6 and model M7. It can be observed that the difference between models M5 and M7 is smaller than the difference between models M7 and M6.

Fig. 20 shows the number of vehicles in the network under demand conditions 4 and 5 for models M5, M6, and M7. The variation in the number of vehicles follows a pattern similar to the mean queue lengths observed for the three models under both demand conditions.

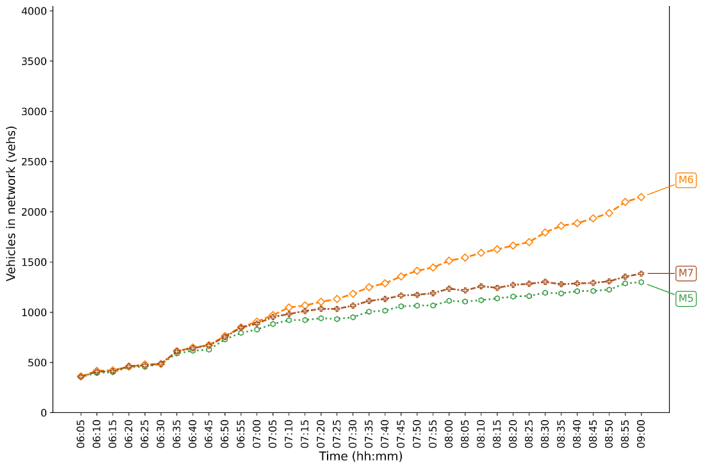
4.6. Empirical analysis under extreme demand conditions

We conducted an empirical analysis of PASCAL model under three additional demand conditions (6, 7, and 8). For these, demand condition 1 was increased by 45 %, 50 %, and 55 %, respectively. The total number of vehicles generated over the 3-hour simulation period was 29,000 under demand condition 6, 30,000 under demand condition 7, and 33,000 under demand condition 8. Fig. 21 shows the number of vehicles loaded in every 15-minute interval under these three demand conditions compared to demand condition 1.

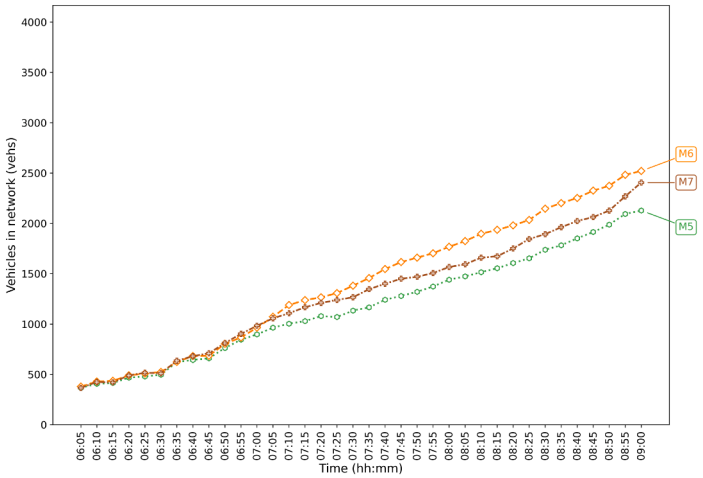
Fig. 22 shows how the number of vehicles in the network varies over time for models M5, M6, and M7 across demand conditions 6, 7, and 8. It was observed that under these extremely congested regimes, many vehicles waited outside the network as part of virtual queues at the entry links.

Across these three demand conditions, the number of vehicles in the network under all three models increases over time, particularly under demand conditions 7 and 8. Under demand condition 6, models M5 and M7 perform significantly better than model M6. The storage availability factors in models M5 and M7 help accommodate the congested regime and maximize throughput much more effectively than model M6. This is further supported by the simulation results, which show that green time wastage for model M6 is significantly higher than for the other two models due to downstream spillback.

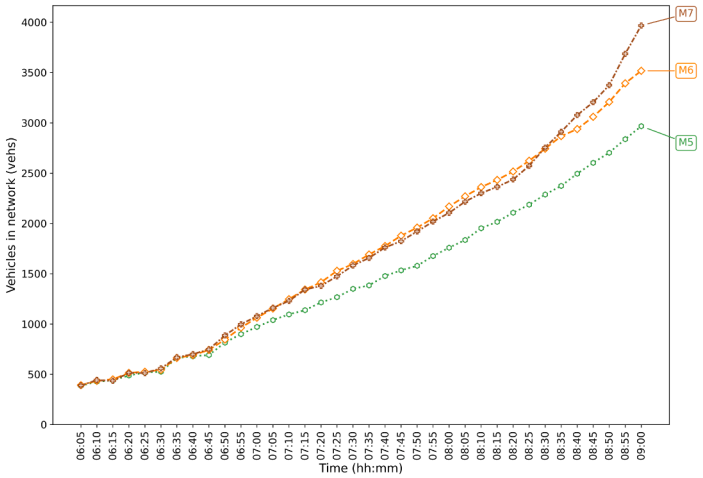
Under demand condition 7, even M5 (PASCAL) was unable to accommodate the demand due to near-spillback conditions on almost all links in the network. The performance deteriorated further under demand condition 8, where the number of vehicles in the network increased rapidly. Under this demand, after 8:30 am, the number of vehicles in the network for model M7 is higher than that for model M6. However, this is due to the substantially larger virtual queue, or the number of vehicles waiting to enter the network at the entry



(a) Demand condition 6



(b) Demand condition 7



(c) Demand condition 8

Fig. 22. Vehicles in the network under demand condition 6 and 7 resulted by model M5, M6, and M7.

links, for model M6 as compared to M7. To examine this, the variation in the number of vehicles waiting to enter under demand condition 8 is shown in Fig. 23.

As shown in Fig. 23, although the number of waiting vehicles starts to grow rapidly after 7:25 am for both models M6 and M7,

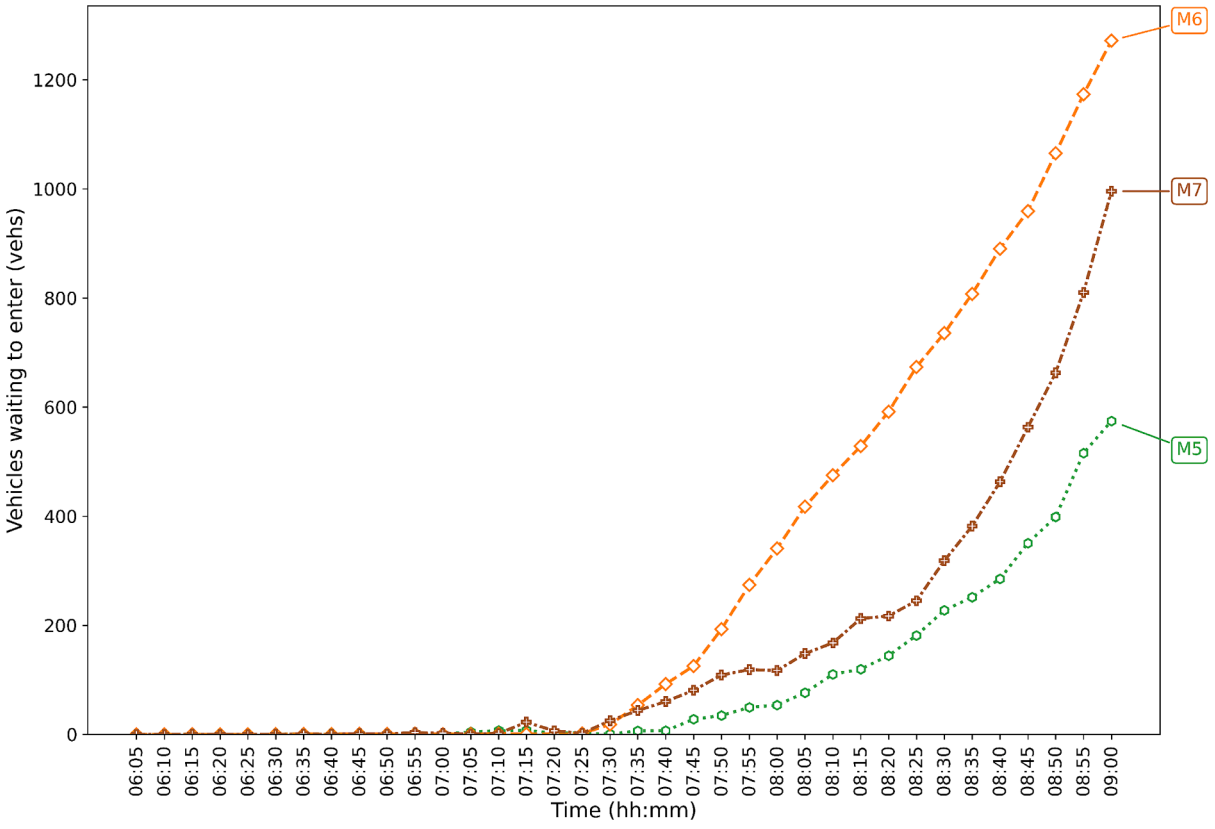


Fig. 23. Number of vehicles waiting to enter the network under demand condition 8.

model M7 reports a significantly lower virtual queue than model M6 over the simulation period. Model M5 exhibits the lowest virtual queue among the three models under demand condition 8. Furthermore, it can be observed that the number of vehicles waiting to enter begins to grow gradually for model M5 only after 7:40 am, representing a delayed onset of spillback at the entry links by nearly 15 min (between 7:25 am and 7:40 am) compared with the other models.

Overall, the analysis suggests that PASCAL can accommodate vehicles in the network up to demand condition 6 with the synthetic demand and control conditions created in these simulation experiments. Beyond this level, PASCAL (model M5) also fails, and virtual queues at the entry links begin to grow rapidly. These higher demand conditions lead to substantial oversaturated conditions across the intersections in the network. Thus, the analysis indicates that the PASCAL pressure function with group-based control maintains stability within the feasible demand region.

4.7. Reproducibility

In this research, PASCAL is implemented using the Aimsun Application Programming Interface (API). Each intersection in the network is controlled by its own Aimsun API file, and all files run in parallel within the Aimsun platform. It was observed that it only takes 0.001 s per intersection, per time step, to compute the signal control decision on a 12th Generation Intel® Core™ i7-1265U (1.80 GHz) processor with 16.0 GB RAM. Thus, even if the intersections were controlled in series rather than in parallel, up to 1,000 intersections could be managed within one second. The benchmark MP control models, which are also implemented using the Aimsun API, exhibit similar computational times for each intersection.

Codes for all the benchmarking models including PASCAL were uploaded in a GitHub repository (https://github.com/tamojitghosh/PASCAL). The benchmarking models can be used for further evaluation and testing under various demand and control conditions.

5. Conclusions

This paper presented PASCAL, a pressure-based signal control algorithm inspired by the max-pressure (MP) framework. PASCAL incorporates a group-based signal control logic into the MP framework and introduces a novel pressure function designed to address practical challenges in urban traffic control, such as limited storage capacity in both candidate lane(s) and downstream link of a movement. The pressure function regulates the occupancy of the candidate lane(s) (referred to as queue length in MP studies) through

two factors, (a) the reduction in storage capacity of the candidate lane(s) and (b) the reduction in storage capacity of the downstream link for a specific movement. PASCAL employs a dynamic time-slot structure, where at the end of each slot a critical movement and a set of critical complementary movements are identified for activation. The green time is allocated based on prevailing traffic conditions, constrained by minimum and maximum green times, with amber and clearance red times provided for safety.

To ensure fairness, PASCAL ensures that all movements are served at least once before any movement is repeated as the critical movement, without relying on a fixed phase sequence. Instead, the model uses a dynamic critical movement selection pool, which is updated at the end of each dynamic time slot to determine the next critical movement. When a movement continues from one time slot to the next, it receives uninterrupted green time, thereby avoiding unnecessary amber and clearance red transitions and preserving the overlapping principle of group-based signal control.

PASCAL was tested in a controlled simulated traffic network consisting of five intersections. Its performance was compared against four MP control models as well as a fixed-time control with SIDRA-optimized parameters. The four MP models were: (a) the acyclic MP control by Varaiya (2013b), (b) the cyclic MP control by Le et al. (2015), (c) the MP with cyclical structure by Levin et al. (2020), and (d) the capacity-aware MP control by Gregoire et al. (2014). Initially, three demand conditions were tested, and various performance measures were reported. PASCAL outperformed all the signal control models across all three conditions. For instance, under a high demand condition (demand condition 3), the aggregated mean delay time per vehicle per kilometer (averaged across all data aggregation intervals over the simulation period for that demand condition) achieved by PASCAL was 50 % lower than SIDRA, and 48 %, 53 %, 45 %, and 31 % lower than the four MP models, respectively.

To further assess its robustness, the pressure functions of Varaiya (2013b) and Gregoire et al. (2014) were embedded into the same group-based control framework used in PASCAL. Under the same three demand conditions, performance measures showed that PASCAL achieved slightly better results. To evaluate the benefit of the PASCAL pressure function under higher congestion levels, two additional increased demand conditions were tested. In these cases, PASCAL significantly outperformed the other two models, even though all three models employed the same group-based signal control logic. This highlights the efficiency of the PASCAL pressure function. All models, including PASCAL, were coded using the Aimsun API architecture, and parameter values were fine tuned for each demand condition.

The proposed signal control algorithm does not explicitly model pedestrian flows or minimum green times required to ensure safe crossings, and it also does not incorporate signal coordination or public transport priority. Future research should extend PASCAL to multimodal contexts and coordinated networks to further enhance adaptability and efficiency. In addition, the current PASCAL formulation does not account for movement-specific (lane-level) storage capacity in downstream links. Explicitly incorporating movement- or lane-specific downstream storage into the pressure formulation is a promising direction for future research.

CRediT authorship contribution statement

Tamojit Ghosh: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lelitha Devi Vanajakshi:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation, Conceptualization. **Md. Mazharul (Shimul) Haque:** Methodology, Conceptualization. **Ashish Bhaskar:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

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Appendix A

The existing MP control studies are summarized in this section. At first, a table of notation (Table A1) is presented and then Table A2 below provides a detailed summary of the existing MP control studies, including phase selection, phase timing criteria for each study, and the corresponding weight function, pressure function, phase identification policy, and green time calculation.

Table A1
Notations and their descriptions.

Notation	Description
w	Weight function
$m \in \Phi$	Any specific movement at an intersection that belongs to a specific predefined phase Φ
t	Discrete time step for studies with incremental green and cycle time for studies with fixed cycle
z_m	Movement m within the candidate link z
z_m^d	Downstream link connected to movement m of the candidate link z
δ_m	Saturation flow rate of movement m
S^*	Optimal decision of which phase to be activated

(*continued on next page*)

Table A1 (*continued*)

Notation	Description
$x_{z_m z_m^d}$	Occupancy, or number of vehicles in the lane(s) of movement m of candidate link z, that are willing to move to the downstream link z_m^d connected to movement m
c_z	Storage capacity of link z
c_∞	Infinite storage capacity or a large number adopted in the study by Gregoire et al. (2014)
β	Positive constant that defines the finite queue and depends on link characteristics
r	Average turning proportion from candidate link to the downstream link
$x^u_{z_m}$	Sum of queue lengths of all the upstream links that feed vehicles to the candidate lane(s)z_m
g_Φ	Green time allocated for phase Φ
L	Total lost time
γ	Positive constant
$\{g_\Phi(t)\}$	Set of green times for all the phases
g_{min}	Minimum green time for each phase
G_{min}	Total minimum green time for all the phases
$\theta_\Phi, \vartheta_\Phi$	Weight parameters of phase Φ that depends on the ratio of number of vehicles in a link to storage capacity of the link, θ and ϑ, respectively
θ, ϑ	Positive constants
q^f	Virtual queue for a specific flow f, where f belongs to a specific origin–destination (OD) pair
$\mathscr{F}$	All flows that pass through a specific pair of candidate and downstream links
q^Δ	Virtual queue for a specific destination Δ
u	Waiting time of front-line vehicles
μ^u	Weight parameter for waiting time
μ^x	Weight parameter for queue length
ξ_{z_m}	Length of link z_m
X	Position of vehicles
φ	Density of traffic
j	Index of an intersection
v	Average vehicle volume of an upstream link directed to a downstream link
λ	Average delay for a turning movement
$\mathscr{T}$	Planning horizon
τ	Index of the planning horizon $\mathscr{T}$
J	Total number of intersections in a network
y	Average travel time at a link
y^f	Free flow travel time at a link
n	Number of lanes in a link
$S^{RL}(t)$	Action state of a reinforcement learning model
k	Cell index when a link is divided into multiple cells
$Var(x)$	Variance of maximum queue length in each link
g_{max}	Maximum green time for a phase
ρ	Density of a cell in a link
ξ_k	Length of a cell k in a link
P_m^{en}	Endogenous pressure for movement m
g_m	Green time for movement m
P_m^{ex}	Exogenous pressure for movement m
$\mathbb{T}$	Total number of time steps elapsed before which each phase must be given green signal
$\mathbb{H}$	A global integer multiplier constant for all the intersections in the network
$\lvert\Phi\rvert$	Total number of phases in an intersection
$\mathbb{t}_\Phi$	Time steps since a specific phase Φ is activated
$\Phi(S(t))$	A specific phase Φ is activated at time step t
z	A specific lane that belongs to a specific movement
$F\lVert x(t)\rVert$	Sum of all the queue lengths
x_m^B	Number of buses waiting at movement m
$s_m(t)$	A binary variable {0, 1} that represents whether movement m is activated at time t
$\widehat{x}$	Predicted number of vehicles in the time step
d_i^b	Delay incurred by vehicle i since the vehicle joined a movement
$\widetilde{g}$	Raw green time allocated to a phase
$s_{\nearrow}(t)$	A binary variable {0, 1} that represents whether pedestrian phase$_{\nearrow}$ is activated at time t
$\varrho_{\nearrow}$	Waiting time of pedestrians for the pedestrian phase$_{\nearrow}$
$\widehat{\varrho}_{\nearrow}$	Maximum threshold waiting time of pedestrians for the pedestrian phase$_{\nearrow}$
$\overline{\mathbb{v}}_{z_m z_m^d}$	Space-mean speed of vehicles at the candidate lane(s) z_m waiting to move to the downstream link z_m^d
$\mathbb{v}^{ffs}$	Free flow speed of vehicles
$o_{z_m z_m^d}$	Average passenger occupancy of the vehicles at z_m intending to move to link z_m^d
$\mathbb{p}$	A protected region within a network that consists of multiple intersections
$\varphi^{\mathbb{p}}$	Average density of the links of the protected region
$\varphi_{cr}^{\mathbb{p}}$	Average critical density of the protected region
$\mathbb{C}$	Coordination weight function (≥ 0)
$\mathbb{c}_m(t)$	Coordination indicator for movement m at time t
$\mathfrak{v}$	Vehicle type
$PCU_\mathfrak{v}$	Passenger car unit of vehicle type $\mathfrak{v}$

Table A2
Comparative overview of max-pressure (MP) control studies. (*The weight and pressure functions, and green time formulations summarized in this table are presented using notation and terminology consistent with the current paper for ease of comparison. In several cases, formulations from the original studies have been simplified or reformulated to highlight their core structure. Readers are referred to the cited publications for the exact original expressions and implementation details.).

Study	Phase selection		Phase timings		Performance measure	Weight function*	Pressure function*	Critical phase identification	Green time*
	Acyclic	Cyclic	Fixed cycle	Incremental green					
Wongpiromsarn et al. (2012)	X			X	Link occupancy	$w_{m\in\Phi}(t) = x_{Z_m}(t) - x_{Z_m^d}(t)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$	NA
Varaiya (2013b)	X			X	Movement occupancy	$w_{m\in\Phi}(t) = x_{Z_m Z_m^d}(t) - x_{Z_m^d}(t)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$	NA
Gregoire et al. (2014)	X			X	Link occupancy	$w_{m\in\Phi}(t) = \min\left(\frac{\frac{x_{Z_m}(t)}{c_\infty} + \left(2 - \frac{c_Z}{c_\infty}\right)\left(\frac{x_{Z_m}(t)}{c_Z}\right)^\vartheta}{1 + \left(\frac{x_{Z_m}(t)}{c_Z}\right)^{\vartheta-1}}, 1\right) - \min\left(\frac{\frac{x_{Z_m^d}(t)}{c_\infty} + \left(2 - \frac{c_{Z^d}}{c_\infty}\right)\left(\frac{x_{Z_m^d}(t)}{c_{Z^d}}\right)^\vartheta}{1 + \left(\frac{x_{Z_m^d}(t)}{c_{Z^d}}\right)^{\vartheta-1}}, 1\right)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$	NA
Xiao et al. (2014)	X			X	Link occupancy	$w_{m\in\Phi}(t) = \widehat{w}_{Z_m Z_m^d}(t) - \sum_{Z_m^{d'}} r_{Z_m^d Z_m^{d'}}\,\widehat{w}_{Z_m^d Z_m^{d'}}(t)$, $\widehat{w}_{Z_m Z_m^d}(t) = \beta_{Z_m Z_m^d} x^u_{Z_m}\frac{x_{Z_m Z_m^d}(t)}{c_{Z_m}}$, $0 < \beta_{Z_m^d Z_m^{d'}} \le \beta_{Z_m^d Z_m^{d'}} r_{Z_m^d Z_m^{d'}}$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$	NA
Kouvelas et al. (2014)		X	X		Link occupancy	$w_{m\in\Phi}(t) = \frac{x_{Z_m}(t)}{c_Z} - \frac{x_{Z_m^d}(t)}{c_{Z^d}}$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	NA	$g_\Phi(t) = \frac{P_\Phi(t)}{\sum_\Phi P_\Phi(t)}(t - L - G_{min}) + g_{min}$
Le et al. (2015)		X	X		Link occupancy	$w_{m\in\Phi}(t) = x_{Z_m}(t) - x_{Z_m^d}(t)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	NA	$g_\Phi(t) = \frac{exp\{\gamma P_\Phi(t)\}}{\sum_\Phi exp\{\gamma P_\Phi(t)\}}t$
Pumir et al. (2015)		X	X		Movement occupancy	$w_{m\in\Phi}(t) = x_{Z_m Z_m^d}(t) - x_{Z_m^d}(t)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	NA	$\{g_\Phi(t)\} = \arg\max_{g_1,\dots,g_{\lvert\Phi\rvert}} \sum_\Phi g_\Phi P_\Phi(t)$, $g_\Phi \ge g_{min}$, $\sum_\Phi g_\Phi \le 1 - \frac{L}{t}$
Xiao et al. (2015b)		X	X		Movement occupancy	$w_{m\in\Phi}(t) = x_{Z_m Z_m^d}(t) - x_{Z_m^d}(t)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	NA	$g_\Phi(t) = \frac{exp\{\gamma P_\Phi(t)\}}{\sum_\Phi exp\{\gamma P_\Phi(t)\}}(t - L - G_{min}) + g_{min}$
Xiao et al. (2015a)		X	X		Movement occupancy	$w_{m\in\Phi}(t) = x_{Z_m Z_m^d}(t) - x_{Z_m^d}(t)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	NA	
Zaidi et al. (2016)	X			X	Virtual queue length based on flow	$w_{m\in\Phi}(t) = \arg\max_{f\in\mathcal{F}}(q^f_{Z_m}(t) - q^f_{Z_m^d}(t))$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$	NA
						$w_{m\in\Phi}(t) = \max_\Delta(q^\Delta_{Z_m}(t) - q^\Delta_{Z_m^d}(t))$			
Wu et al. (2018)	X			X	Waiting time of front row vehicles	$w_{m\in\Phi}(t) = u_{Z_m Z_m^d}(t)$, $w_{m\in\Phi}(t) = \mu^u u_{Z_m Z_m^d}(t) + \mu^x x_{Z_m Z_m^d}(t)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$	NA
Anderson et al. (2018)		X	X		Movement occupancy	$w_{m\in\Phi}(t) = x_{Z_m Z_m^d}(t) - x_{Z_m^d}(t)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	NA	$\{g_\Phi(t)\} = \arg\max_{g_1,\dots,g_{\lvert\Phi\rvert}} \sum_\Phi g_\Phi P_\Phi(t)$, $g_\Phi \ge g_{min}$, $\sum_\Phi g_\Phi \le 1 - \frac{L}{t}$
Li and Jabari (2019)	X			X	Link-based vehicle position	$w_{m\in\Phi}(t) = \left\vartheta_{Z_m Z_m^d}\int_0^{\xi_{Z_m}}\left\lvert\frac{X}{\xi_{Z_m}}\right\rvert\varphi_{Z_m Z_m^d}(t)dx - \int_0^{\xi_{Z_m^d}}\left\lvert\frac{\xi_{Z_m^d} - X}{\xi_{Z_m^d}}\right\rvert\vartheta_{Z_m^d Z_m^{d'}} r_{Z_m^d Z_m^{d'}}\varphi_{Z_m^d Z_m^{d'}}(t)dx$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$	NA
Wei et al., 2019	X			X	Movement occupancy	$w_{m\in\Phi}(t) = \frac{x_{Z_m}(t)}{c_Z} - \frac{x_{Z_m^d}(t)}{c_{Z^d}}$	$P_j(t) = \left\lvert\sum_\Phi\sum_m w_{m\in\Phi}(t)\,\delta_m(t)\right\rvert$	$S^{RL}(t) = \arg\max_j -P_j(t)$	NA

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Table A2 (*continued*)

Dixit et al. (2020)		X	X		Delay	$w_{m\in\Phi}(t) = \frac{v_{z_m z_m^d} \lambda_{z_m z_m^d}(t)}{\delta_m(t)}$	$P_\Phi(t) = \max_{m\in\Phi}\{w_{m\in\Phi}(t)\}$	NA	$g_\Phi(t) = \frac{P_\Phi(t)}{\sum_\Phi P_\Phi(t)}(t - L)$
Levin et al. (2020)		X		X	Movement occupancy	$w_{m\in\Phi}(t) = x_{z_m z_m^d}(t) - x_{z_m^d}(t)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	NA	$g_\Phi(t) = max\frac{1}{\mathcal{T}}\sum_{\tau=0}^{\mathcal{T}-1}(1 - L(t + \tau))\,P_\Phi(t)$
Ma et al. (2021)		X	X		Link occupancy	$w_{m\in\Phi}(t) = x_{z_m}(t) - x_{z_m^d}(t)$	$P_j(t) = \sum_j\sum_\Phi\sum_m w_{m\in\Phi}(t)\,\delta_m($	NA	$g_\Phi(t) = \min(P_j(t), Var(x))$ s.t. $g_{min} \le g_\Phi(t) \le g_{max}$
Mercader et al. (2020)		X	X		Link-based travel time	$w_{m\in\Phi}(t) = \frac{y_{z_m}(t)}{y^f_{z_m}} - \sum_{z_m^d} r_{z_m z_m^d}\frac{y_{z_m^d}(t)}{y^f_{z_m^d}}$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	NA	$g_\Phi(t) = \frac{P_\Phi(t)}{\sum_\Phi P_\Phi(t)}(t - L - G_{min}) + g_{min}$
Bai and Bai (2021)	X			X	Link occupancy	$w_{m\in\Phi}(t) = x_{z_m z_m^d}(t) - \frac{x_{z_m^d}(t)}{n_{z_m^d}}$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$	NA
Boukerche et al. (2021)	X			X	Movement occupancy	$w_{m\in\Phi}(t) = x_{z_m z_m^d}(t) - x_{z_m^d}(t)$	$P_j(t) = \sum_\Phi\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^r(t) = \arg\max_j -P_j(t)$	
Li et al. (2021)	X			X	Link occupancy	$w_{m\in\Phi}(t) = \sum_{k\in z_m}\rho_k(t)\xi_k - \sum_{k\in z_m^d}\rho_k(t)\xi_k$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$	NA
Yu et al. (2021)		X	X		Movement occupancy and storage capacity	$p_m^{en} = \frac{x_{z_m^u z_m}(g_m) - x_{z_m z_m^d}(g_m)}{g_m}$ $p_m^{ex} = \frac{c_{z^d} - x_{z_m^d z_m^{d'}}}{g_m}$			
Liu and Gayah (2022)	X			X	Delay of stopped vehicles	$w_{m\in\Phi}(t) = \sum_{t'=1}^{t}\left[x^s_{z_m z_m^d}(t') - x^s_{z_m^d}(t')\right]$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$	NA
Robbennolt et al. (2022)		X		X	Movement occupancy	$w_{m\in\Phi}(t) = x_{z_m z_m^d}(t) - x_{z_m^d}(t)$	$P_\Phi(t) = \sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t) = \arg\max_\Phi P_\Phi(t)$ s.t. $\mathbb{T} = \mathbb{H} * \lvert\Phi\rvert$ $\exists\Phi\lvert\mathbb{t}_\Phi \ge \mathbb{T} \to \begin{cases}\Phi(S(t)) = 1\\ S^*(t)\end{cases}$	

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Table A2 (*continued*)

Zoabi and Haddad (2022)		X	X		Lane-level travel time delay	$w_{z\in\Phi}(t) = \sqrt[\gamma]{\frac{y_{z\in m}(t)-y^f_{z\in m}}{y^f_{z\in m}}} - \sum_{z^d\in m} r_{z^d}\sqrt[\gamma]{\frac{y_{z^d\in m}(t)-y^f_{z^d\in m}}{y^f_{z^d\in m}}}$	$P_\Phi(t)=\sum_z w_{z\in\Phi}(t)\,\delta_z(t)$	NA	$g_\Phi(t) = \frac{P_\Phi(t)}{\sum_\Phi P_\Phi(t)}(t-L-G_{min})+g_{min}$
Wang et al. (2022)	X			X	Movement occupancy	$w_{m\in\Phi}(t)=x_{z_m z^d_m}(t)-x_{z^d_m}(t)$	$P_\Phi(t)=\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S(t)=\arg\max_\Phi P_\Phi(t)$ $S^*(t) = \begin{cases} S(t-1) & S(t)<S(t-1)+F\\ S(t) & Otherwise\end{cases}$	NA
Xu et al. (2022)	X			X	Bus and private vehicle movement wise occupancy	$w_{m\in\Phi}(t)=x_{z_m z^d_m}(t)-x_{z^d_m}(t)$	$P_\Phi(t)=\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t)=\arg\max_\Phi P_\Phi(t)$ s.t. $\sum_{m\in\Phi} x^B_m(t)[\sum_{m\in\Phi} s_m(t)x^B_m(t)-1]\geq 0$	NA
Barman and Levin (2023)	X			X	Movement occupancy	$w_{m\in\Phi}(t)=x_{z_m z^d_m}(t)-x_{z^d_m}(t)$	$P_\Phi(t)=\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S(t)=\arg\max_\Phi P_\Phi(t)$	NA
Faqir et al. (2023)	X			X	Movement occupancy	$w_{m\in\Phi}(t)=\hat{x}_{z_m z^d_m}(t)-\hat{x}_{z^d_m}(t)$	$P_\Phi(t)=\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t)=\arg\max_\Phi P_\Phi(t)$	NA
Liu and Gayah (2023)	X			X	Total vehicle delay	$w_{m\in\Phi}(t) = \sum_i d^b_{i,z_m z^d_m} - \sum_{z^d_m, z^{d'}_m}\left(r_{z^d_m z^{d'}_m}\sum_{t'=1}^{t} d_{z^d_m z^{d'}_m}(t')\right)$	$P_\Phi(t)=\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t)=\arg\max_\Phi P_\Phi(t)$	NA
Tsitsokas et al. (2023)		X	X		Link occupancy	$w_{m\in\Phi}(t)=\frac{x_{z_m}(t)}{c_z}-\frac{x_{z^d_m}(t)}{c_{z^d}}$	$P_\Phi(t)=\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	NA	$\tilde{g}_\Phi(t)=\frac{P_\Phi(t)}{\sum_\Phi P_\Phi(t)}(t-L)$ minimize$(\tilde{g}_\Phi-g_\Phi)^2$ $\sum_\Phi^{g_\Phi} g_\Phi + L = t$
Xu et al. (2024)	X			X	Vehicle and Pedestrian occupancy	$w_{m\in\Phi}(t)=x_{z_m z^d_m}(t)-x_{z^d_m}(t)$	$P_\Phi(t)=\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t)=\arg\max_\Phi P_\Phi(t)$ s.t. $\left(1-s_p(t)\right)\left(\varrho_p(t)-\hat{\varrho}_p(t)\right)\leq 0$	NA
								$s_p(t)\leq 1-s_m(t)\chi^p_m$	
Ahmed et al. (2024)	X			X	Movement occupancy and space mean speed	$w_{m\in\Phi}(t)=x_{z_m z^d_m}(t)\left(1+\frac{\bar{\mathbb{V}}_{z_m z^d_m}}{\mathbb{V}^{ffs}}\right)-x_{z^d_m}(t)\left(1-\frac{\bar{\mathbb{V}}_{z^d_m z^{d'}_m}}{\mathbb{V}^{ffs}}\right)$	$P_\Phi(t)=\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t)=\arg\max_\Phi P_\Phi(t)$	NA
Ahmed et al. (2024)	X			X	Movement occupancy and average passenger occupancy of each vehicle	$w_{m\in\Phi}(t)=o_{z_m z^d_m}(t)\left[x_{z_m z^d_m}(t)-x_{z^d_m}(t)\right]$	$P_\Phi(t)=\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t)=\arg\max_\Phi P_\Phi(t)$	NA
Liu and Gayah (2024)	X			X	Delay of stopped vehicles and density of links	$w_{m\in\Phi}(t) = \begin{cases}\sum_{t'=1}^{t}\left[x^s_{z_m z^d_m}(t)-x^s_{z^d_m}(t)\right]-\mathbb{F}(t) & \varphi^{\mathbb{P}}(t)>\varphi^{\mathbb{P}}_{cr}(t)\\ \sum_{t'=1}^{t}\left[x^s_{z_m z^d_m}(t)-x^s_{z^d_m}(t)\right] & otherwise\end{cases}$	$P_\Phi(t)=\sum_m w_{m\in\Phi}(t)\,\delta_m(t)$	$S^*(t)=\arg\max_\Phi P_\Phi(t)$	NA
Zoabi and Haddad (2024)	X			X	Movement occupancy	$w_{m\in\Phi}(t) = \frac{x_{z_m z^d_m}(t)}{c_{z_m z^d_m}/n_{z_m z^d_m}} - \sum_{z^d_m}\frac{r_{z^d_m z^{d'}_m}\,x_{z^d_m z^{d'}_m}(t)}{c_{z^d_m z^{d'}_m}/n_{z^d_m z^{d'}_m}}$	$P_\Phi(t)=\sum_m \frac{w_{m\in\Phi}(t)\delta_m(t)}{n_m}$	$S^*(t)=\arg\max_\Phi P_\Phi(t)$	NA
Xu et al. (2024)	X			X	Movement Occupancy	$w_{m\in\Phi}(t)=x_{z_m z^d_m}(t)-x_{z^d_m}(t)$	$P_\Phi(t)=\left[\sum_m w_{m\in\Phi}(t)\,\delta_m(t) + \mathbb{C}_m\mathbb{c}_m(t)\right]$	$S^*(t)=\arg\max_\Phi P_\Phi(t)$	NA
Agarwal et al. (2024)		X	X		Link Occupancy	$w_{m\in\Phi}(t)=x_{z_m}(t)-x_{z^d_m}(t)$	$P_{\Phi,\mathfrak{v}}(t)=\sum_{m,\mathfrak{v}}[w_{m\in\Phi}(t)]_{\mathfrak{v}}PCU_{\mathfrak{v}}$	NA	NA

Appendix B

A detailed description of the traffic movements, lane configurations and link wise traffic volumes for the five intersections considered for evaluation purpose in this study is presented below:

North side intersection: Fig. B1(a) shows the north side intersection, which has four slip lanes for left turning movements, all of which are unsignalized. The through movements in the northbound and southbound directions each with three lanes, and each right turn with a single lane. In contrast, the eastbound and westbound directions each have two through lanes, with two lanes dedicated to right turns on the westbound approach and one on the eastbound. The demand pattern indicates higher volumes on the northbound and southbound links compared to the eastbound and westbound links. In particular, the westbound link's demand is almost double that of the eastbound link. All the movements at this intersection lead directly to exit links, except for Southbound Through (SB Th), Eastbound Right (EB RT), and Westbound Left Turn (WB LT).

South side intersection: Fig. B1 (b) shows the south side intersection with slip lanes for the eastbound and westbound left turns, which are unsignalized. Each northbound and southbound approach has four lanes, with one lane each for left turns and right turns, and two centre lanes for through movements. Eastbound and westbound approaches have two lanes each: one dedicated through lane and one shared lane for through and right turn movements. As with the north side intersection, most movements here lead to exits except for Northbound Through (NB Th), Eastbound Left Turn (EB LT), and Westbound Right Turn (WB RT). The demand pattern shows that southbound demand is highest, followed by northbound, while eastbound and westbound demands remain lower and are equal to each other.

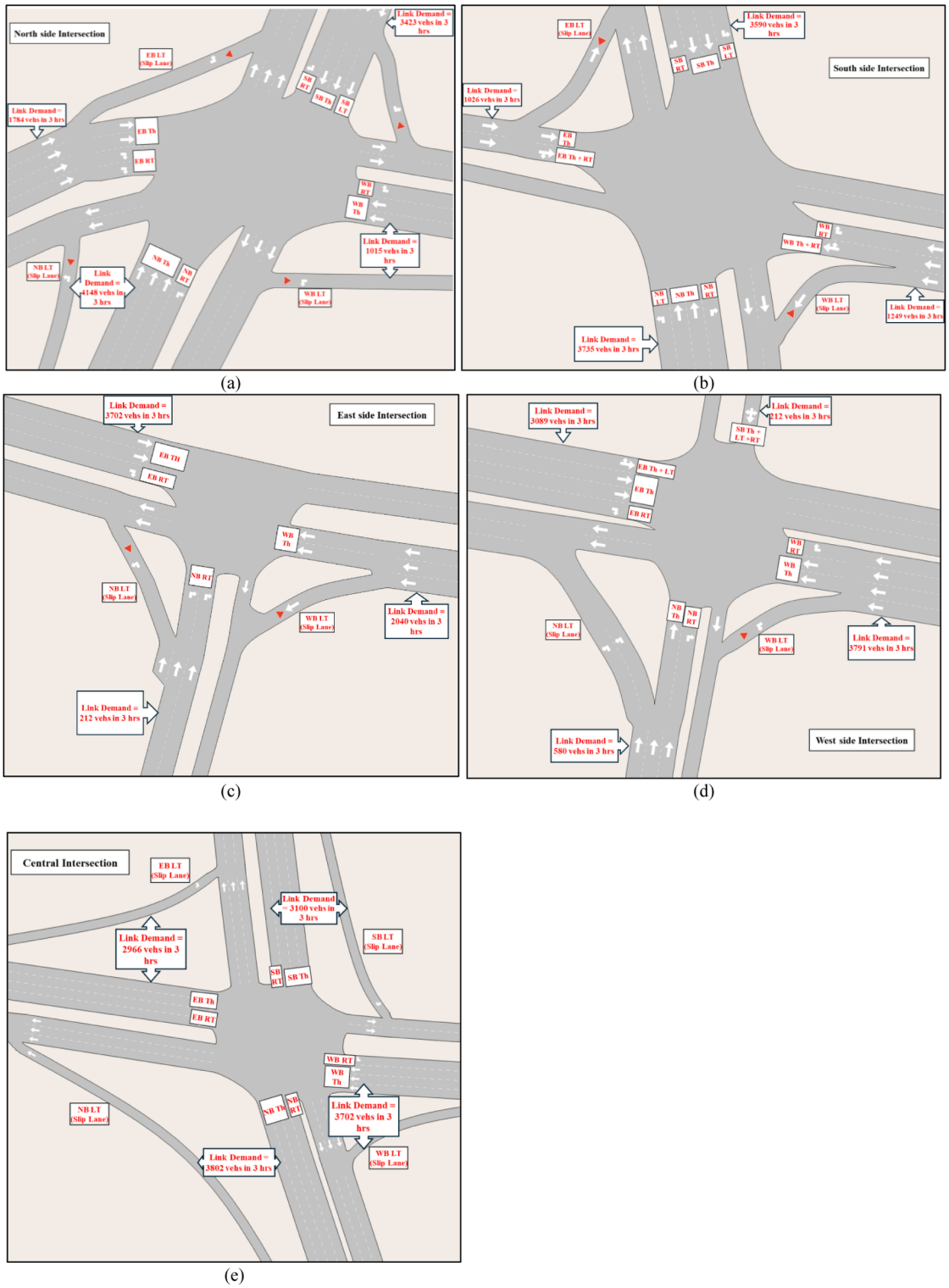


Fig. B1. Layout and link wise demand for all five intersections

East side intersection: Fig. B1 (c) shows the east side intersection, which is a three-legged configuration where the majority of demand comes from the east or west approaches. The northbound and westbound links each have a slip lane for left turns. The northbound approach includes two lanes for right turns, while the eastbound approach has two lanes for through traffic and one

dedicated right turn lane. The westbound approach offers two lanes for through movements. In this case, the highest demand arrives from the east, whereas the northbound link has very low demand because it serves a smaller residential community. Overall, eastbound traffic volumes are moderate, and lower than that on the westbound link.

West side intersection: The west side intersection is four-legged (Fig. B1 (d)). The westbound link has a slip lane for left turns, and the northbound link has two slip lanes for left turns, all unsignalized. Westbound traffic has two lanes for through movement and one dedicated right-turn lane. The eastbound link has two lanes dedicated to through movement, one lane dedicated to right turns, and a leftmost lane shared between through and left turning movements. Northbound traffic has one lane for right turning and one through lane, while the southbound link has only one lane shared among through, left turn, and right turn movements. Demand is slightly higher on the westbound approach than eastbound, but both remain significantly busier than the northbound and southbound approaches.

Central intersection: The central intersection closely resembles the north side intersection in layout, with one key difference: the westbound approach has three through lanes and one right-turn lane (Fig. B1 (e)). The southbound and westbound links have nearly the same (and slightly higher) demands, while the northbound and eastbound links have lower volumes, with northbound demand being just a bit higher than eastbound.

Data availability

Data will be made available on request.

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